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On journalistic responsibility: a note from our faculty advisor

If you have ever had a captive audience eager to learn from you, you have had the responsibility. When scientists publish their work, they are presenting the results and the interpretation of the results in all honesty. The results come from experimentation or observation, and they are reproducible, the critical hallmark for others' accepting the data in the results. In contrast, the interpretation of the results is a matter of discussion, as interpretation comes from the individual's own knowledge base and understanding of the results.

So, too in journalism, including the Berkeley Scientific Journal, the scientific papers are reviewed, revised and published with the understanding that the results are as claimed. The interpretation can sometimes be open to, well, interpretation. Journalism and science have serious communication responsibilities. A journalist is a storyteller whose currency is non-fiction, and whose responsibility is to share the non-fiction in a manner understood by the reader or listener.

A journalist does not share "fake news," nor report unsubstantiated claims as reality. Indeed a journalist has a responsibility to expose "fake news" for the distraction and misdirection it causes. A scientist who writes and speaks to the public becomes a journalist who helps the non-scientist interpret scientific information.

For the audience that is listening to or reading your words, you have a responsibility to them. Communicate clearly and honestly, directly and without exaggeration, respectfully and concisely - truthfully. If you are "reporting," be clear about your sources and evaluate their veracity. If you are sharing your own opinion, make it clear that you are presenting an interpretation of events, results, behaviors, and your opinion is based on your own reasoning, although there may be facts and reports whose impact you are interpreting.

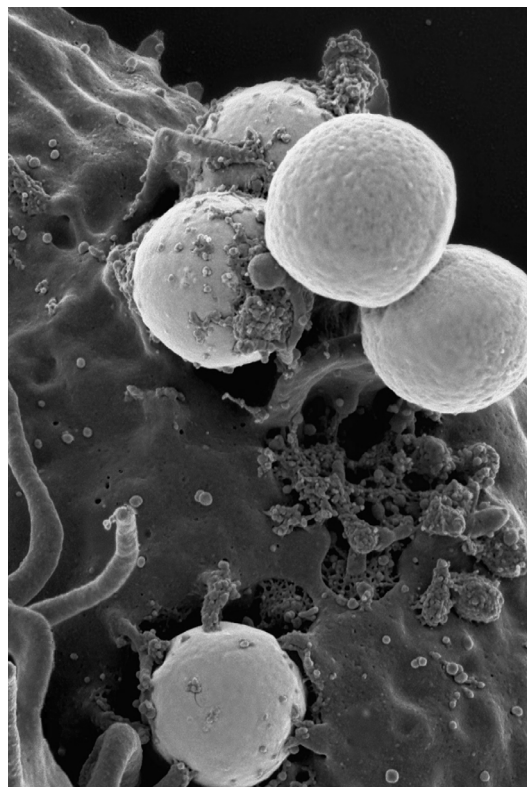
We are all responsible for sharing substantiated information; we are all responsible for exposing claims that lie outside the truth. The journalists of the Berkeley Scientific Journal understand this responsibility and trust that you will too.

-Dr. Caroline Kane, **faculty advisor to BSJ**

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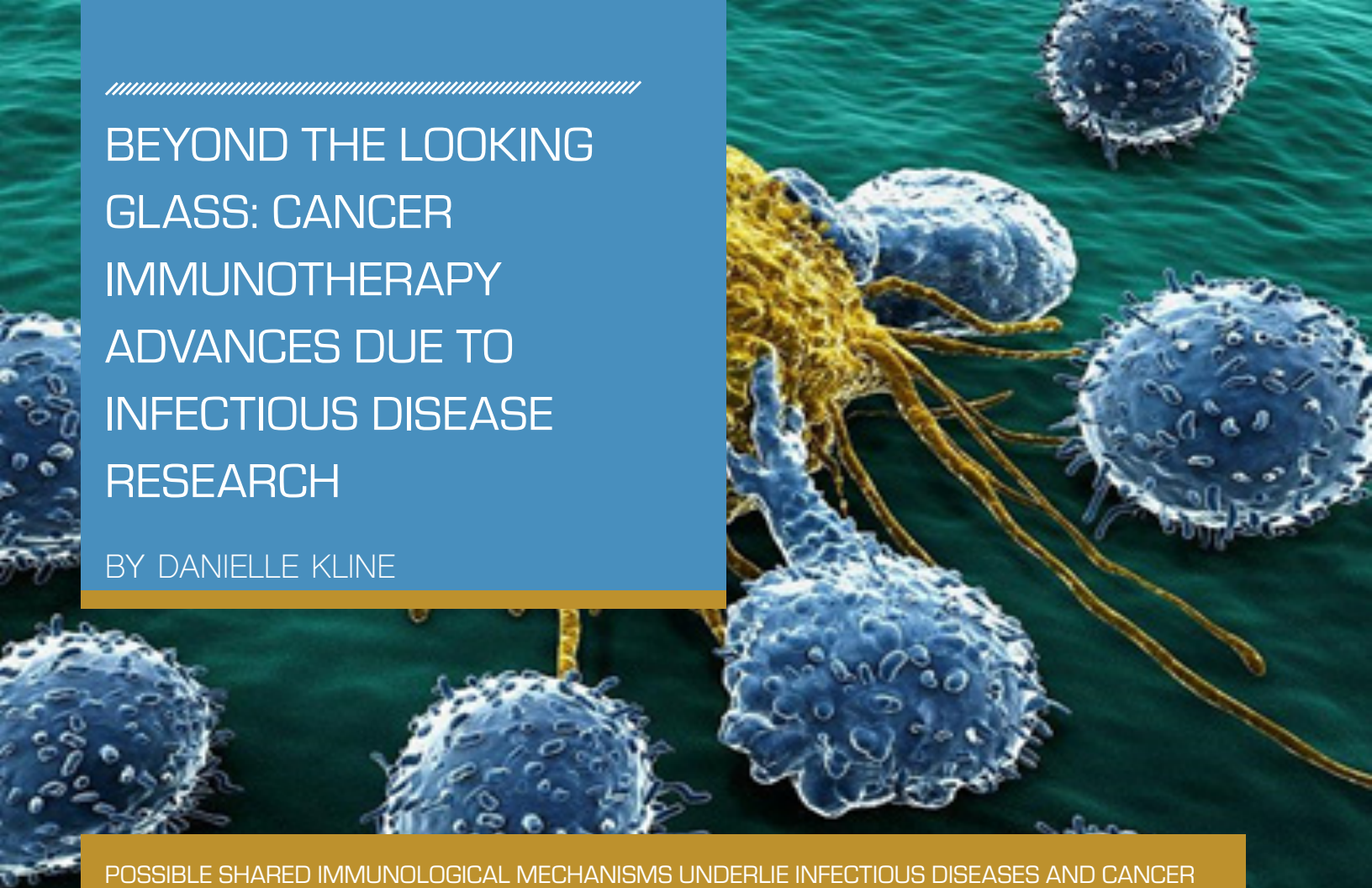
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A microscopic image showing several blue, spherical cells with spiky surfaces, likely immune cells or bacteria, against a green background. Some cells have long, thin, yellowish structures extending from them.

BEYOND THE LOOKING GLASS: CANCER IMMUNOTHERAPY ADVANCES DUE TO INFECTIOUS DISEASE RESEARCH

BY DANIELLE KLINE

POSSIBLE SHARED IMMUNOLOGICAL MECHANISMS UNDERLIE INFECTIOUS DISEASES AND CANCER

Why does the model created for vaccine development work for a wide range of infectious diseases but fall short when it comes to cancer? Until recently, it has been widely held that cancer has little in common with infectious diseases, and that its treatment and prevention should therefore be of a different standard. David Raulet, Laurent Coscoy, Daniel Portnoy and Russell Vance all believe that this has the potential to be a notion of the past. These scientists founded the Immunotherapeutics and Vaccine Research Initiative (IVRI) on the idea that communication between the fields of cancer research and immunology research can spark revolutions in both fields, ultimately translating to innovative new cancer treatments.¹¹ Looking for a new treatment plan for either cancer or infectious diseases is arduous because each subtype comes with a unique set of virulence factors that must be overcome. However, the researchers of IVRI believe that there are enough similarities

between the two fields to bridge this gap.

Cancer evolves rapidly in response to its environment in order to evade our immune response, and since tumors are composed of our own cells, it is more difficult for the immune system to attack the cancer. Our immune system is able to identify most pathogens by pathogen-associated molecular patterns, or PAMPs. A variety of molecules can act as PAMPs, from the lipopolysaccharides on the outer membrane of bacteria to the flagella that allow bacteria to move. Since PAMPs have molecular features that our own cells do not possess, this allows our immune cells' receptors to identify and bind them, and signal for immune response.⁷ Abnormalities in our own cells are sometime referred to as pseudo-PAMPs and trigger similar immune responses.

Tumors wreak havoc wherever they are present, killing neighboring cells and creating genomically unstable progeny. This is how our immune system is able to find tumors.¹¹ This genomic instability, coupled

with the PAMP-like scenarios previously mentioned, causes an immune system reflex, initiating tissue repair and healing responses through inflammation. Too often, however, inflammation becomes chronic and can lead to cellular damage, proliferation, and formation of new blood vessels which can help the cancer spread. According to Dr. Russell Vance, "[a]n interesting way to think of cancer is as a chronic infectious disease—the consequence of the unresolved presence and growth of a 'foreign' body."¹⁰ With this concept in mind, immunotherapy researchers hope to enhance our own immune response to prevent the spread of malignant cells.

IVRI's faculty director, Dr. David Raulet, hopes to accomplish this goal using Natural Killer (NK) cells. NK cells use specific methods to recognize cancerous cells, one of which is by upregulating proteins through activated stress pathways. In Dr. Raulet's words: "[t]he cells essentially say 'Kill me! I'm stressed out!' The NK cell

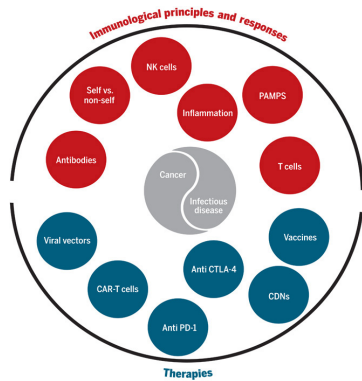


Figure 1: A figure demonstrating exactly how much overlap there is between infectious diseases and cancer. The image includes some of the shared immunological principles, responses, and therapies.¹⁴

kills that cell and then calls on the rest of the immune system to come help out.²⁹⁸

One critical component in the recognition of pathogens by the immune system is the major histocompatibility complex (MHC). The MHC is an assembly of proteins that displays fragments of pathogens on its surface for targeting by the immune system. Some cells undergoing malignant transformations may downregulate their MHCs to reduce immune recognition and confer selection of such cells; however, they can be combated through the strategy of increasing the number of modified NK cells in the bloodstream, which have been mutated to allow their attack on endogenous can-

cerous cells. This is where immunotherapy comes in. Immunotherapy is the treatment (or prevention) of disease using supplements to our own immune system. Most immunotherapy overlooks any and all preconceived notions about how to approach cancer and instead goes back to the basics: improving our immune response to eliminate abnormalities within our bodies. If NK cells from either the patient or a donor are injected into the patient's blood, it is possible to increase the number of NK cells in the bloodstream and enable these new NK cells to have tumor-attacking capabilities.⁴

Perhaps the approach that is most indicative of the leaps and bounds made in cancer immunology is the *Listeria*-based

vaccine. *Listeria monocytogenes* is a pathogen that elicits a strong immune response. To generate a vaccine strain, the virus is attenuated with mutations so that its immunogenic capabilities are retained without any toxicity. The patient's immune system begins rapid production of T-cells to combat the potential infection and this response can be engineered to target the types of cells found in a tumor. Although this research is still in the clinical trial stage, the *Listeria* vaccine has been shown to cure mice with late-stage pancreatic cancer. Researchers are now wondering if the *Listeria* vaccine could be applied to other pathogens, such as HIV or tularemia, that we have yet to develop sound treatments for.¹¹

Until just a few years ago, cancer and infectious diseases were considered to be two completely different entities. However, recent findings have led to a paradigm shift that has revolutionized this notion and opened doors in the field of cancer immunotherapy. As Dr. Russell Vance admits, "We were surprised to find the amazing number of similarities between our immune responses [to cancer and infectious diseases]. Historically, tumor biology and infectious disease biology were considered very different fields, and now we are able to appreciate all of the overlap."¹⁰ These innovative approaches have the potential to not only revolutionize the field of immunotherapy but also offer new treatment options to cancer patients worldwide. The more we are able to learn about the nature of tumors and how to treat cancer, the closer we come to eventually curing the disease.

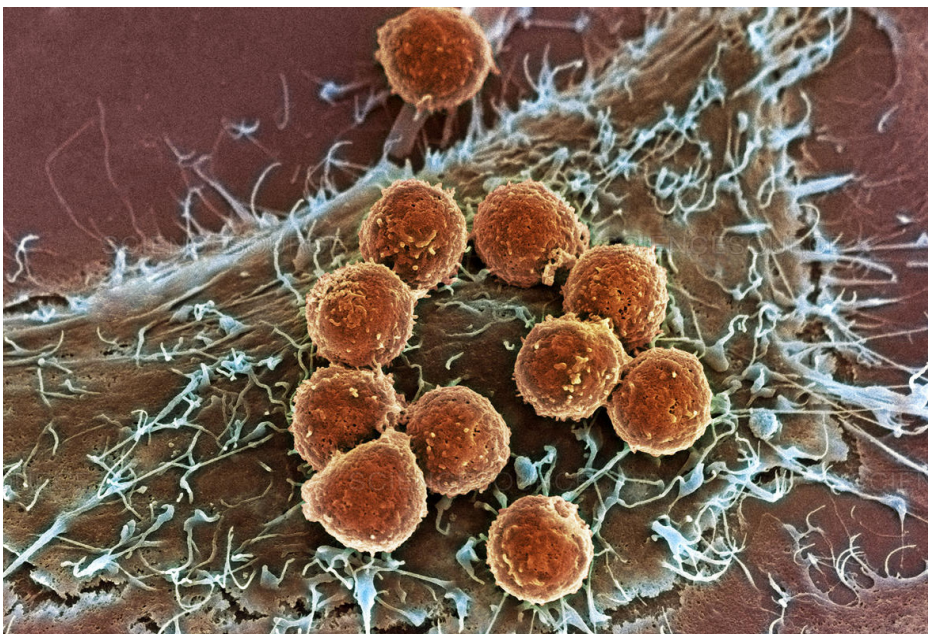


Figure 2: A T-lymphocyte latches onto a malignant tumor. The T-cell identifies a pseudo-PAMP on the surface of the tumor and binds to it, triggering an immune response.¹³

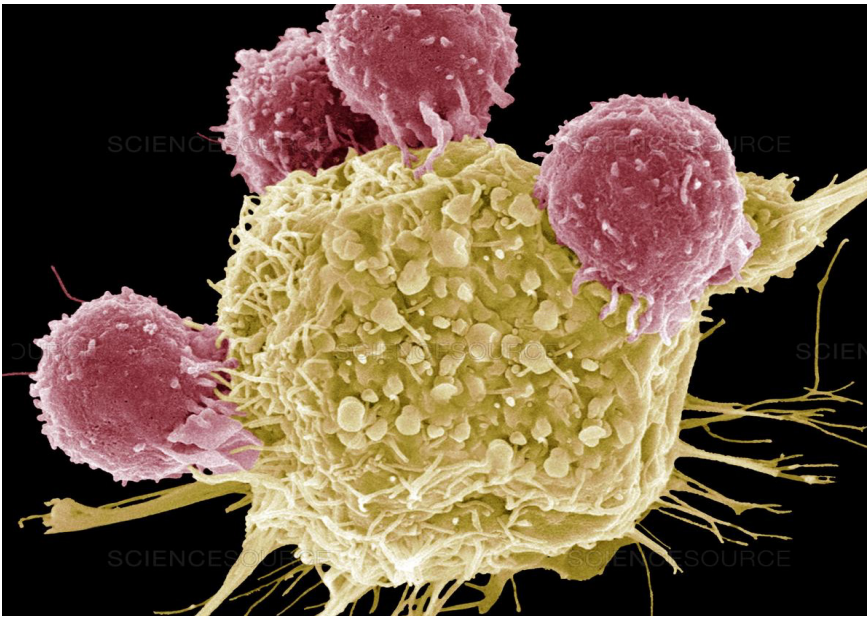


Figure 3: T-lymphocytes latching onto a malignant cell.¹⁵

"An interesting way to think of cancer is as a chronic infectious disease—the consequence of the unresolved presence and growth of a 'foreign' body."

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REFERENCES

1. Brockstedt, D. G., Gledlin, M. A., Leong, M. L., Bahjat, K. S., Gao, Y., Luckett, W., Liu, W., Cook, D. N., Portnoy, D. A., & Dubensky, T.W. (2004, September 21). Listeria-based cancer vaccines that segregate immunogenicity from toxicity. PNAS, 101 (38). Retrieved from <http://www.pnas.org/content/101/38/13832>.
2. Corrales, L., McWhirter, S.M., Dubensky, T.W., & Gajewski, T.F. (2016 July). The host STING pathway at the interface of cancer and immunity. The Journal of Clinical Investigation, 126 (7). Retrieved from <https://>

www.ncbi.nlm.nih.gov/pmc/articles/PMC4922692/.

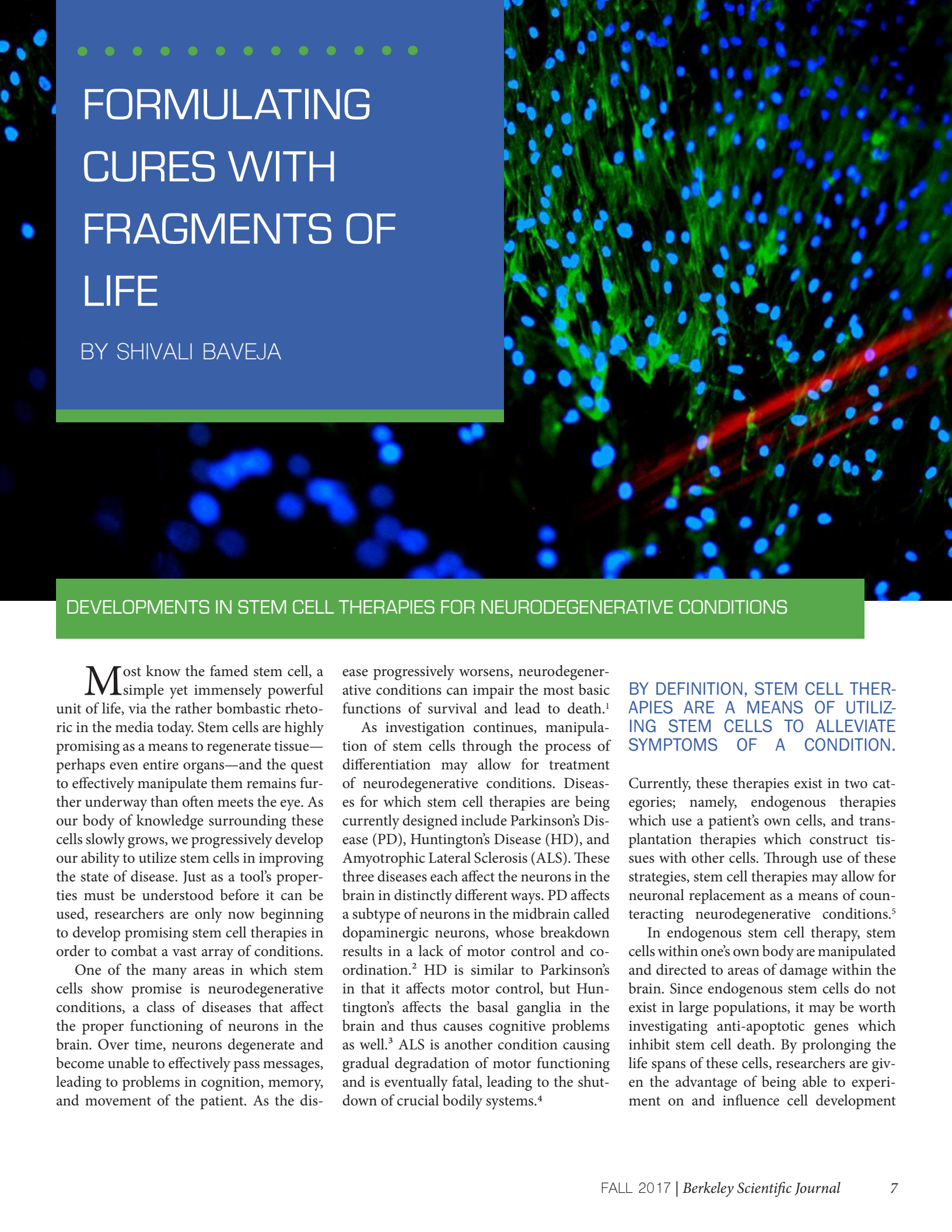
3. Draper, S. J., & Heeney, J. L. (2010, January). Viruses as vaccine vectors for infectious diseases and cancer. Nature, 8. Retrieved from <https://www.nature.com/nrmicro/journal/v8/n1/full/nrmicro2240.html>.
4. Gillerey, C., Huntington, N. D., & Smyth, M. J. (2016 September). Targeting natural killer cells in cancer immunotherapy. Nature Immunology, 17 (9). Retrieved from <https://www.nature.com/ni/journal/v17/n9/full/ni.3518.html>.
5. Hanahan, D., & Weinberg, R. A. (2011, March 4). Hallmarks of Cancer: The Next Generation. Cell, 144. Retrieved from <https://tinyurl.com/yaxo3yhq>.
6. Melief, C. J. M., van Hall, T., Arens, R., Ossendorp, F., & van der Burg, S. H. (2015, September). Therapeutic cancer vaccines. The Journal of Clinical Investigation, 125 (9). Retrieved from <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4588240/>.
7. Rakoff-Nahoum, S., & Medzhitov, R. (2009 January). Toll-like receptors and cancer. Nature Reviews, 9. Retrieved from <https://www.nature.com/nrc/journal/v9/n1/full/nrc2541.html>
8. Raulet, D. H. (2017, October 13). Personal Interview.
9. Tsai, H. F., & Hsu, P. N. (2017). Cancer immunotherapy by targeting immune

checkpoints: mechanism of T cell dysfunction in cancer immunity and new therapeutic targets. Journal of Biomedical Science, 24 (35). Retrieved from <https://tinyurl.com/y7uwpwqf>.

10. Vance, R. E. (2017, October 11). Personal Interview.
11. Vance, R. E., Eichberg, M. J., Portnoy, D. A., & Raulet, D. H. (2017, January 13). Listening to each other: Infectious disease and cancer immunology. Science Immunology. Retrieved from <http://immunology.sciencemag.org/content/2/7/eaai9339.full>.
12. Vogt, P.K. (2012, September). Retroviral oncogenes: a historical primer. Nature Reviews, 12. Retrieved from <https://www.nature.com/nrc/journal/v12/n9/full/nrc3320.html>.

IMAGE SOURCES

13. Colored scanning electron micrograph (SEM) of T lymphocyte cells. (2015). T lymphocytes and cancer cell, SEM. Science Source. Retrieved from <http://images.sciencesource.com/p/11171031/T-lymphocytes-and-cancer-cell-SF5825.html>.
14. Vance, R. E., Eichberg, M. J., Portnoy, D. A., & Raulet, D. H. (2017, January 13). Listening to each other: Infectious disease and cancer immunology. Science Immunology. Retrieved from <http://immunology.sciencemag.org/content/2/7/eaai9339.full>.
15. Immunotherapeutics and Vaccine Research Initiative Home Page. (2017). IVRI. Retrieved from <https://www.sciencesource.com/p/11171031/T-lymphocytes-and-cancer-cell-SF5825.html>.
16. Stern, V. (2015, March 23). Today's Cancer Research Pioneers. Medscape. Retrieved from https://www.medscape.com/viewarticle/841813_2.



FORMULATING CURES WITH FRAGMENTS OF LIFE

BY SHIVALI BAVEJA

DEVELOPMENTS IN STEM CELL THERAPIES FOR NEURODEGENERATIVE CONDITIONS

Most know the famed stem cell, a simple yet immensely powerful unit of life, via the rather bombastic rhetoric in the media today. Stem cells are highly promising as a means to regenerate tissue—perhaps even entire organs—and the quest to effectively manipulate them remains further underway than often meets the eye. As our body of knowledge surrounding these cells slowly grows, we progressively develop our ability to utilize stem cells in improving the state of disease. Just as a tool's properties must be understood before it can be used, researchers are only now beginning to develop promising stem cell therapies in order to combat a vast array of conditions.

One of the many areas in which stem cells show promise is neurodegenerative conditions, a class of diseases that affect the proper functioning of neurons in the brain. Over time, neurons degenerate and become unable to effectively pass messages, leading to problems in cognition, memory, and movement of the patient. As the dis-

ease progressively worsens, neurodegenerative conditions can impair the most basic functions of survival and lead to death.¹

As investigation continues, manipulation of stem cells through the process of differentiation may allow for treatment of neurodegenerative conditions. Diseases for which stem cell therapies are being currently designed include Parkinson's Disease (PD), Huntington's Disease (HD), and Amyotrophic Lateral Sclerosis (ALS). These three diseases each affect the neurons in the brain in distinctly different ways. PD affects a subtype of neurons in the midbrain called dopaminergic neurons, whose breakdown results in a lack of motor control and coordination.² HD is similar to Parkinson's in that it affects motor control, but Huntington's affects the basal ganglia in the brain and thus causes cognitive problems as well.³ ALS is another condition causing gradual degradation of motor functioning and is eventually fatal, leading to the shutdown of crucial bodily systems.⁴

BY DEFINITION, STEM CELL THERAPIES ARE A MEANS OF UTILIZING STEM CELLS TO ALLEVIATE SYMPTOMS OF A CONDITION.

Currently, these therapies exist in two categories; namely, endogenous therapies which use a patient's own cells, and transplantation therapies which construct tissues with other cells. Through use of these strategies, stem cell therapies may allow for neuronal replacement as a means of counteracting neurodegenerative conditions.⁵

In endogenous stem cell therapy, stem cells within one's own body are manipulated and directed to areas of damage within the brain. Since endogenous stem cells do not exist in large populations, it may be worth investigating anti-apoptotic genes which inhibit stem cell death. By prolonging the life spans of these cells, researchers are given the advantage of being able to experiment on and influence cell development

for longer periods of time. Understanding these genes may allow researchers to prevent cell death and generate more efficient endogenous stem cell therapies.⁵ Another area of exploration is induced pluripotent stem cells (iPSCs), adult cells which have been converted back into an undifferentiated state. The process of differentiation is what causes a stem cell to develop into a particular cellular subtype which then retains a specific function. By understanding how cells transition between differentiated and undifferentiated states, more types of endogenous stem cell populations can be developed for use in therapies.⁶

ON THE OTHER HAND, TRANSPLANTATION STEM CELL THERAPIES DIFFERENTIATE EXTERNAL STEM CELLS AND DEVELOP THEM INTO TISSUES MADE TO MATCH THE RECIPIENT.

Thus far, research has been done on transplantation of various types of stem cells, including mesenchymal and neural stem cells (MSCs and NSCs respectively). MSC have been seen to be very effective at shielding themselves from the immune system, thus decreasing chances of the host rejecting them once transplanted. In fact, studies using fetal MSCs in models for HD saw a decrease in the rate of the condition's development and onset.⁷ In trials like these, the transplanted stem cells increased the rate of growth and thus the rate of recovery of damaged neurons. Further research on NSCs has shown similar improvements. When tested in mice, transplanted NSCs resulted in reduced symptoms of Sandhoff Disease, a disease similar to ALS in its effects on motor neurons. Continued trials have shown improvement in symptoms for PD

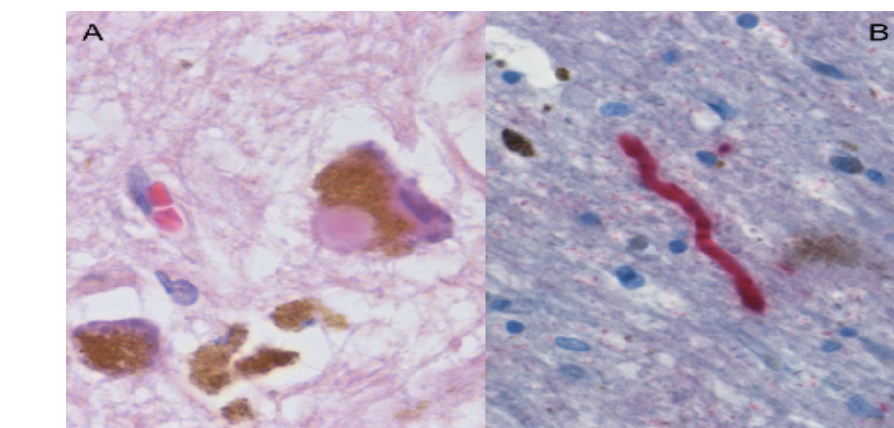


Figure 1: Lewy Bodies (a) and Lewy Neurites (b) are indicators of neurodegeneration particularly as seen in Parkinson's Disease.¹⁶

patients with NSC transplantations as well.⁸

Despite the small-scale successes of transplantation therapies, running clinical trials for them has proven to be difficult. Ethical concerns surrounding the source of stem cells are often difficult to overcome for potential participants in stem cells therapies. Increasing awareness about stem cell research—the vast majority of which does not involve embryonic stem cells at all—would greatly increase the feasibility of clinical trials.

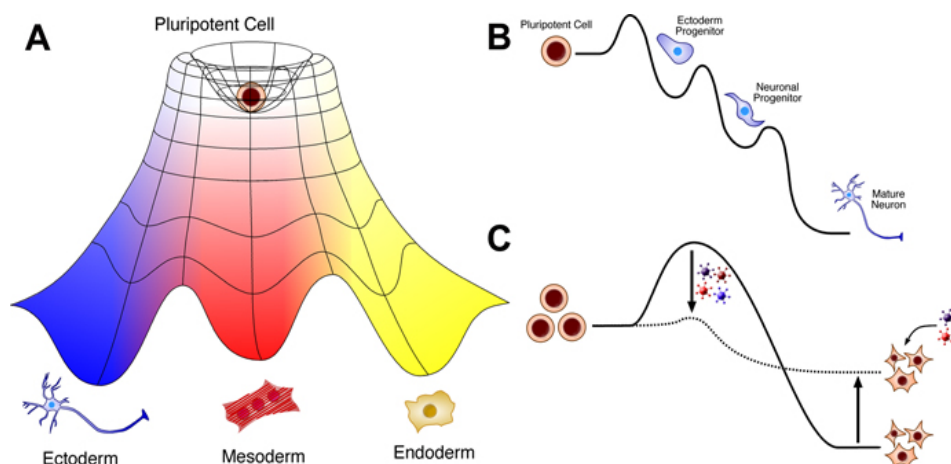
Scientists have broken down the areas in which growth is necessary for stem cell trials to develop. First, within animal models, there is a need for greater proof of the potential of integrating stem cells into the larger neuronal system. Second, a means to expand the lifespan of transplanted stem cells would also be greatly

beneficial in tissue testing and development. Third, for application to trials, further analysis of how different patients are affected by individual therapies would allow for more personalized trials to be run.¹⁰

BEYOND THE GENERIC AREAS OF DEVELOPMENT, THERE IS SCOPE TO EXPERIMENT WITH DIFFERENT TYPES OF STEM CELLS.

As bone marrow-derived stem cells continue to be explored, other types of cells being investigated include endothelial and neural stem cells. Amniotic fluid stem cells have also been seen to be very promising, although the procedure of amniocentesis necessary to harvest the cells can be rather risky. Finding means to extract these cells may also be an avenue to explore further.¹¹

Figure 2: (a) Pluripotent stem cells can differentiate into one of three larger categories of which the ectoderm includes neuronal subtypes. (b) There exist several distinct progenitor states for cells as they differentiate from their initial pluripotent states to neurons.¹⁷



“RESEARCHERS ARE ALSO GROWING CELLS IN 3D STRUCTURES TO BETTER MIMIC THE ACTUAL BODILY CONDITIONS, ALLOWING DEVELOPMENT OF TRANSPLANTATION AND ENDOGENOUS THERAPIES.”

As researchers work with these cells, tools for differentiating them are being created and discovered. Genetic manipulation is one way this can be accomplished, as RNA sequencing for singular cells currently allows us to control neuronal development. This technique allows us to observe and manipulate the biological makeup of a cell through gene transcription, a process which indirectly determines a cell's identity and behavior. One gene being targeted via RNA sequencing is WNT7A, a gene controlling progenitor cell replication, which if successfully modified would allow us to build frameworks for cell implantation.¹² Physical tools can also be used to construct structures for cells. An example of this is a cellular bio-bridge, which is being used to guide the development and migration of endogenous stem cells.¹³ Researchers are also growing cells in 3D structures to better mimic the actual bodily conditions, allowing development of transplantation and endogenous therapies.¹⁴

Developments in stem cell therapy are burgeoning as different types of cells and tools emerge from the accumulating body of research. The use of both transplantation and endogenous therapies provide paths to bettering the lives of those facing neurodegenerative conditions. Although we continue to learn more about the stem cell, there is ever more left to explore in order to develop effective therapies such that these fragments of life can better our lives.

REFERENCES

1. “What Is Neurodegenerative Disease.” JPNP, www.neurodegenerationresearch.eu/about/what/.
2. “Parkinson’s Disease.” Mayo Clinic, Mayo Foundation for Medical Education and Research, 7 July 2015, www.mayoclinic.org/diseases-conditions/parkinsons-disease/basics/definition/con-20028488.
3. “Huntington’s Disease.” Mayo Clinic, Mayo Foundation for Medical Education and Research, 13 June 2017, www.mayoclinic.org/diseases-conditions/huntingtons-disease/symptoms-causes/syc-20356117.
4. “Amyotrophic Lateral Sclerosis.” Mayo Clinic, Mayo Foundation for Medical Education and Research, 12 May 2017, www.mayoclinic.org/diseases-conditions/amyotrophic-lateral-sclerosis/symptoms-causes/syc-20354022.
5. Kishk, N., & Abokrysh, N. (2011). Stem Cell in Neurological Disorders. Stem Cells in Clinic and Research. doi:10.5772/21408
6. Lindvall, O., & Kokaia, Z. (2010). Stem cells in human neurodegenerative disorders — time for clinical translation? Journal of Clinical Investigation, 120(1), 29-40. doi:10.1172/jci40543.
7. Scuteri, A. (2012). Treatment of Neurodegenerative Pathologies Using Undifferentiated Mesenchymal Stem Cells. Stem Cells and Cancer Stem Cells, Volume 6, 185-195. doi:10.1007/978-94-007-2993-3_16.
8. Lee, J., Jeyakumar, M., Gonzalez, R., Takahashi, H., Lee, P., Baek, R. C., . . . Snyder, E. Y. (2007). Stem cells act through multiple mechanisms to benefit mice with neurodegenerative metabolic disease. Nature Medicine, 13(4), 439-447. doi:10.1038/nm1548.
9. Aled, J., Delavaran, H., Lindval, O., Norrvig, B., Kokaia, Z., & Lindgern, A. (2017). Attitudes to Stem Cell Therapy Among Ischemic Stroke Survivors in the Lund Stroke Recovery Study. Stem Cells and Development, 26(8). doi:10.1089/scd.2016.0343.
10. Lindvall, O., Kokaia, Z., & Martinez-Serrano, A. (2004). Stem cell therapy for human neurodegenerative disorders—how to make it work. Nature Medicine, 10(7). doi:10.1038/nm1064.
11. Corey, S., Ghanekar, S., Sokol, J., Zhang, J., & Borlongan, C. (2017). An update on stem cell therapy for neurological disorders: cell death pathways as therapeutic targets. Chinese Neurological Journal, 3(4). doi:10.1186/s41016-016-0071-2.
12. Toledo, E. M., Gyllborg, D., & Arenas, E. (2017). Translation of WNT developmental programs into stem cell replacement strategies for the treatment of Parkinsons disease. British Journal of Pharmacology. doi:10.1111/bph.13871
13. Napoli, E., & Borlongan, C. V. (2017). Cell Therapy in Parkinsons Disease: Host Brain Repair Machinery Gets a Boost From Stem Cell Grafts. Stem Cells, 35(6), 1443-1445. doi:10.1002/stem.2636.
14. Correia, C., Koshkin, A., Duarte, P., Carido, M., Lima, P., Teixeira, A., . . . Serra, M. (2017). Novel Strategies for Generation and Hypothermic Storage of the Human Pluripotent Stem Cell-Derived Cardiomyocytes for Cell Therapy Applications. Cytotherapy, 19(5). http://dx.doi.org/10.1016/j.jcyt.2017.03.040ylene terephthalate) Science 1196-11.

IMAGE SOURCES

15. https://www.flickr.com/photos/cambridgeuniversity-engineering/14310429488/in/photostream/.
16. By Werner CJ, Heyny-von Haussen R., Mall G., Wolf S. - Werner CJ, Heyny-von Haussen R., Mall G., Wolf S. Proteome analysis of human substantia nigra in Parkinson’s disease.. Proteome Sci. 6, 8. 2008. doi:10.1186/1477-5956-6-8. PMID 18275612., CC BY 2.0, https://commons.wikimedia.org/w/index.php?curid=3946973.
17. By Rodolfa, K.T., Inducing pluripotency (September 30, 2008), StemBook, ed. The Stem Cell Research Community, StemBook, doi/10.3824/stem-book.1.22.1.

COMPUTING TO THE TARGET: ACCELERATING ORPHAN DRUG DISCOVERY

BY KARA JIA

When most people think about pharmaceutical drugs, the first ones that come to mind are usually popular over-the-counter (OTC) drugs such as Tylenol, Advil, and Zyrtec. These medications are seemingly omnipresent; they can be found in ordinary places, from local pharmacies to convenience stores. While OTC drugs are among the most accessible medications to all consumers, they constitute only the tip of the iceberg of all existing and potential pharmaceutical drugs. Prescription drugs are required to treat most medical conditions apart from the common cold. New drugs are constantly under development for major diseases. Yet while treatments for rare diseases inflicting a much smaller population may be in the stages of drug development, many more remain non-existent.

What exactly sets rare diseases—also known as orphan diseases—apart from common ones? Orphan diseases affect a small subset of a population. In United States, such conditions typically affect less than 1 out of 200,000 people, and currently there are about 7,000 such diseases affecting approximately 25 million patients.⁶ These medical conditions generally lack attention from pharmaceutical companies as they do not yield large commercial success. Government efforts to increase economic incentives led to the passing of the Orphan Drug Act (ODA) in 1983, which encourages development and marketing of drugs for rare diseases. Among the main incentives for orphan drug research and development are grants, research design support, and a seven-year exclusive orphan drug marketing.⁴

Since the enactment of ODA, the number of orphan drugs produced has significantly increased. There is a perpetual need for orphan drug development as about 250 new rare diseases are reported each year.⁴ At present, drugs for rare cancer subtypes represent the majority (31%) of approved orphan drugs.⁶ Fabrazyme, an enzyme replacement therapy, is an example of an approved orphan drug used to treat Fabry's disease, an ultra-rare X-linked genetic disorder caused by deficiency in the enzyme alpha-galactosidase A. Under this disease, gastrointestinal symptoms, ophthalmological symptoms, and neuropathic pain manifest in

early childhood due to the accumulation of glycosphingolipids—lipids that are part of the cell membrane and crucial for cell-cell interactions—in different organs.¹⁰

Developing drugs such as Fabrazyme entails a time-consuming, costly, and difficult process. Basic research first generates the necessary data that initiates and fuels the discovery phase of drug development. During the discovery phase, potential candidate molecules—or leads—are selected. A lead is a chemical compound with pharmacological or biological activity that is potentially therapeutically useful. Lead compounds enter a pre-clinical validation stage; selected candidate leads are then permitted to enter clinical phases. If the clinical trials are successful, the FDA filing process for the manufacturing of an orphan drug begins.

Drug discovery is initiated for diseases that do not yet have suitable medical products. Identifying a target molecule is the start of a lengthy process, which involves a concerted effort between academic research institutions and the pharmaceutical industry. The drug discovery process can be divided into the following stages: initial target identification and validation, assay development, high throughput screening, hit identification, lead optimization, and selection of a lead molecule for clinical development. Only 10% of potential leads actually make it to clinical trials.¹ Because there is a high attrition rate for compounds entering the clinical phase, novel techniques could significantly help optimize the drug discovery process. A failed drug not only impedes the goal of bringing effective treatment to patients but also leads to higher financial consequences for pharmaceutical companies, thus slowing down the whole drug development process. With an average of \$2.6 billion invested in developing a marketable drug, finding new techniques that would help reform the drug discovery pipeline thus remains a major priority. The efficiency of computers has made in silico methods an indis-



“While treatments for rare diseases inflicting a much smaller population may be in the stages of drug development, many more remain nonexistent.”

pensable tool for expediting the process of finding promising and novel drug leads.⁵

Compared to traditional methods such as high-throughput screening (HTS), computer-aided drug design (CADD) tools have been shown to be more efficient and complementary.² Two main types of CADD are structure-based drug design (SBDD) and ligand-based drug design (LBDD). SBDD is a method focusing on predicting and analyzing 3D structures of target proteins (i.e. finding a molecule with the best fit in an active site). Prediction of target structure is based on the assumption that “proteins with similar sequences have similar structures.”² Molecular dynamics simulations are used in SBDD to reveal the various pathways for ligands to interact with target proteins as well as the different possible target conformations.⁷ Accumulation of biological data makes SBDD studies particularly favorable.⁹ On the other

hand, LBDD is used to analyze possible ligands that interact with the target of interest to determine whether the small molecule increases or decreases target activity. It is used when the protein structure cannot be experimentally or computationally determined and relies on information about known active ligands.⁷

Other than finding novel drug leads, repositioning approved drugs is another possible approach to orphan drug development. Repurposing FDA-approved drugs is relatively inexpensive compared to developing de novo drugs.⁸ Duloxetine is a serotonin and norepinephrine reuptake inhibitor that has been approved to treat depression. Since serotonin and norepinephrine were found to be key neurotransmitters in fibromyalgia (a central nervous system disorder) and chronic musculoskeletal pain, the duloxetine pathway was successfully repositioned

to treat these disorders.⁶ One way to discover novel properties of approved drugs is through the use of computational methods; these insights could then be used to develop drugs for orphan diseases.⁸ For this purpose, machine learning and text mining are two promising computational approaches.

One proposed machine learning model creates algorithms devised to help predict unknown drug-disease associations.⁸ This method integrates genome, phenome, and chemical structure information into a computational framework that extracts a drug similarity matrix and a disease similarity matrix. Insights gained from this approach could potentially aid drug repositioning by revealing novel drug indications.⁹

Taking a more indirect approach, text mining can be used to extract particular terms and phrases from electronically-stored literature. Given a set biological ontology, text mining could then be used

“The efficiency of computational methods has made in silico methods an indispensable tool for expediting the process of finding promising novel drug leads.”



Figure 1: Fabrazyme is an approved orphan drug used to treat Fabry's disease.¹¹

for retrieval of relevant information on a particular drug in literature which could yield novel indications for existing drugs.⁸ Different institutions produce data organized in different ways; thus, text mining could also prove useful in standardizing heterogeneous data.³

Combined with economic incentives and efficient computational methods, more opportunities are created for orphan drug discovery as well as general drug development. While each computational method has its own set of limitations and challenges, the drug discovery process benefits from any new, potentially useful insights that can minimize drug discovery time for the many orphan diseases which are, as of yet, still incurable.

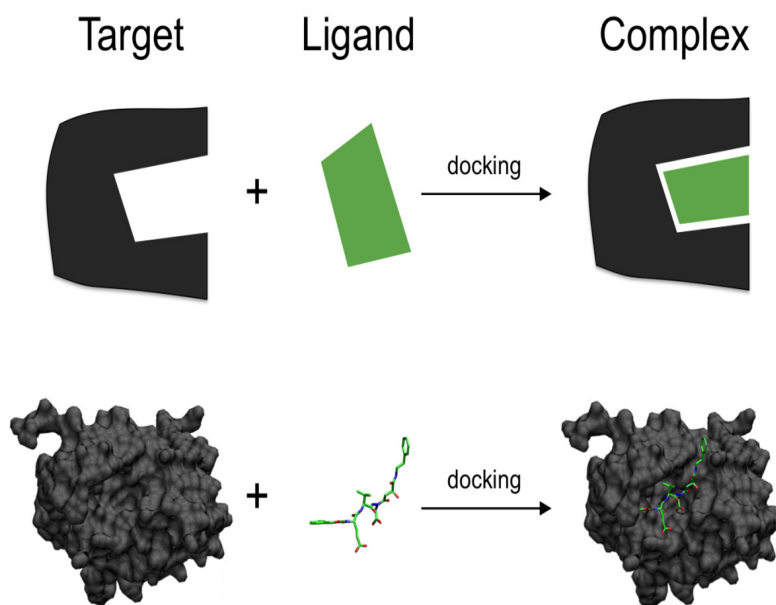
REFERENCES

1. Hughes, J., Rees, S., Kalindjian, S., & Philpott, K. (2011). Principles of early drug discovery. *British Journal of Pharmacology*, 162(6), 1239–1249.
2. Sliwoski, G., Kothiwale, S., Meiler, J., & Lowe, E. W. (2014, January). *Computational Methods in Drug Discovery*.
3. Yao, L., Evans, J. A., & Rzhetsky, A. (2010, April). Novel opportunities for computational biology and sociology in drug discovery.
4. Seoane-Vazquez, E., Rodriguez-Monguió, R., Szeinbach, S. L., & Visaria, J. (2008, December 16). Incentives for orphan drug research and development in the United States.
5. Molecular dynamics-driven drug discovery: leaping forward with confidence. (2016, November 25).
6. Li, Y. Y., & Jones, S. J. (2012, March 30). Drug repositioning for personalized medicine.
7. Leelananda, S. P., & Lindert, S. (2016, December 12). *Computational methods in drug discovery*.
8. Muthyala, Ramaiah. Orphan/rare drug discovery through drug repositioning. (2011, November 04).
9. Li, J., Zheng, S., Chen, B., Butte, A. J., Swamidass, S. J., & Lu, Z. (2015, March 31). A survey of current trends in computational drug repositioning.
10. Karpman, D., & Höglund, P. (2016, October 13). Orphan drug policies and use in pediatric nephrology.

IMAGE SOURCES

11. <https://upload.wikimedia.org/wikipedia/commons/e/e2/FabrazymeProductPicture.jpg>.
12. https://upload.wikimedia.org/wikipedia/commons/thumb/9/97/Docking_representation_2.png/1200px-Docking_representation_2.png.

Figure 2: The docking of a ligand to a protein target leading to the formation of a protein-ligand complex.¹²





DECODING THE NEURAL CODING PROBLEM

BY SANIKA GANESH

In every human is a philosopher that ponders the meaning of life. Modern neuroscience seeks to provide some insight to our existential anxiety by investigating the neural basis of consciousness: how do we, as human beings, consciously perceive the world around us? This inquiry frames what neuroscientists have termed the neural coding problem. The neural coding problem investigates how the brain codes and synthesizes information to regulate an organism's behavior.

Neural coding describes how neurons in our brain and nervous system process stimuli from the environment. Neurons communicate with each other using electrical signals, called action potentials, and chemical signals, called neurotransmitters. Dramatic spikes of electrical activity characterize action potentials, and these spikes can be analyzed to understand how an organism reacts to stimuli.⁹ How an organism interprets and behaves in response to stimuli can change over time. Neuroplasticity is the way in which an organism's brain changes and adapts through experience, "to learn and remember patterns in the sensory world, to refine movements, to predict and obtain reward, and to recover function after injury." Regions of the brain, like the neocortex, are centers for the assimilation of information in regards to learning.⁴

Scientists today have a variety of tools to analyze neural activity in the brain. Two of the most important methods are electrode-based techniques and functional brain imaging. Electrode-based techniques use microelectrodes to measure spikes of activity, or action potentials, in one or a few neurons at a time. An analysis of this activity determines how a neuron is tuned—how it responds to a specific stimulus—emphasizing the encoding aspect of sensation. Unlike electrode-based techniques, functional brain imaging does not require the insertion of microelectrodes into the brain and instead uses advanced technology to perform scans of the brain. Experiments typically utilize functional brain imaging to indirectly examine the activity of thousands or billions of neurons by measuring changes in blood flow.⁹ Because the human nervous system is very complicated, scientists study simple organisms like the fruit fly and the rodent to understand the fundamental ways in which neurons function and to test the limits of our knowledge about this phenomenon.

Experiments with rodents illustrate the precise nature of sensory processing. For example, rodents have evolved to detect texture and roughness by analyzing the friction their whiskers feel as the whiskers drag and slip upon contact with a surface.

"These higher-level brain processes come closer to tracking our conscious thoughts—potentially allowing us to 'mind-read.'"

BY RECODING SPIKES, RESEARCHERS CAN TRACK HOW RODENTS CALCULATE FRICTION.

Rodents can use this detailed sensory information to carefully discriminate the texture of different surfaces.⁸ As evidenced by even this relatively ordinary task, the nervous system is powerfully exact.

Studies of the fruit fly *Drosophila* demonstrate that such precise sensory information is used for elaborate decision-making. In order to navigate flight, *Drosophila* collects information about its surroundings using visual sensory processing. The fly also keeps track of its own motion and position in space. These cues are somehow consolidated in the brain in order to allow for decision-making about the course of its flight.¹⁰ Though scientists can track neural responses to these stimuli, exactly how different types of information are integrated within the mind of an organism is unknown. The organization of sensory data is broadly explained by the property of neuroplasticity: the fruit fly uses neuroplasticity to make order of the overwhelming amount of sensory and motor information that it experiences during flight. The fly's decision-making process uses patterns of information to learn and respond accordingly.

The diverse ways in which organisms like humans and *Drosophila* react to the same stimuli establish scientific grounds for the subjectivity of life experience. For instance, neuroplasticity enables gustation through "taste coding." *Drosophila* organizes sensory information into different channels that are in some way desegregated to develop learned associations for particular taste molecules.⁷ Research regarding gustation in *Drosophila* has also revealed that these fruit flies can taste water.³ Interestingly, humans do not have the gustatory sensors to taste water. Because organisms have distinct ways of understanding the same stimuli, how they experience the environment also varies drastically.

The unique experiences of pain and itch also involve neuroplasticity. Whether itch and pain have separate pathways is still debated, though it is understood that both experiences are interconnected. Both pain and itch can be transient (acute) or chronic. Acute pain is usually a protective reflex in

response to life-threatening conditions, while acute itch can be stimulated by a variety of sensations, including pain.^{1,2} These reflexes help organisms be aware of their surroundings. On the other hand, "chronic itch, like chronic pain, can occur without injury or disease, serves no apparent biological purpose and has no recognizable endpoint," according to an article by the scientific journal *Cell*.² Mechanisms of both acute and chronic pain or itch are facilitated by neuroplasticity, but only acute pain or itch is deemed particularly beneficial.¹ The nervous system may amplify signals like that of pain or itch to enhance perception of the original stimulus. Chronic pain and itch exemplify the limits of a neuron's ability to code information and productively make sense of the environment, suggesting that perception is extremely nuanced and, perhaps, slightly imperfect. The brain produces perception—every organism's perception of its environment is unique, and therefore, every organism's reality is subjective.

Using functional brain imaging, researchers have developed methods of detecting what a human brain visually perceives about an object and its action

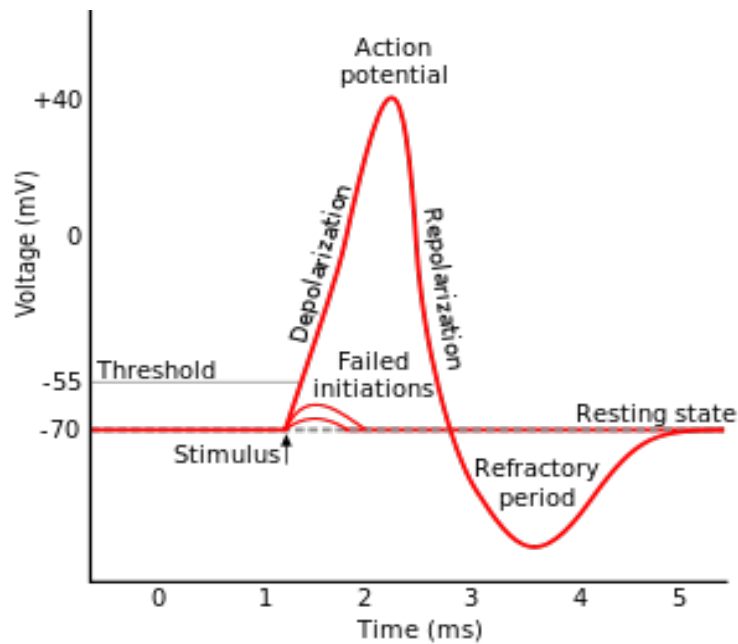


Figure 1: A stimulus that induces electrical activity past a critical threshold launches an action potential. The peak on this graph represents an action potential as a single spike of electrical activity. Action potentials propagate across individual neurons, while neurotransmitters transmit these nerve impulses between neurons.

"The brain produces perception—every organism's perception of its environment is unique, and therefore, every organism's reality is subjective."

and the relationship between such objects.

DECODING THE BRAIN SHOWS HOW NEUROPLASTICITY WORKS THROUGH PATTERNS, OR IN THIS CASE, CATEGORIES.

Researchers can use this decoding technique to guess what movie a subject is watching. These higher-level brain processes come closer to tracking our conscious thoughts—potentially allowing us to ‘mind-read.’ For this reason, research in this field may be controversial: how will such techniques and technology be utilized if fully developed?

Scientists today certainly have a better understanding of encoding and decoding than in the past—when the brain was a complete mystery—but the middle ground of this exploration remains uncharted. How sensory information comes into awareness

could very well explain consciousness. Whether humans will eventually conceive the complexity of their own consciousness is something that is left to be discovered, and how this knowledge could transform our existence is an even greater question.

REFERENCES

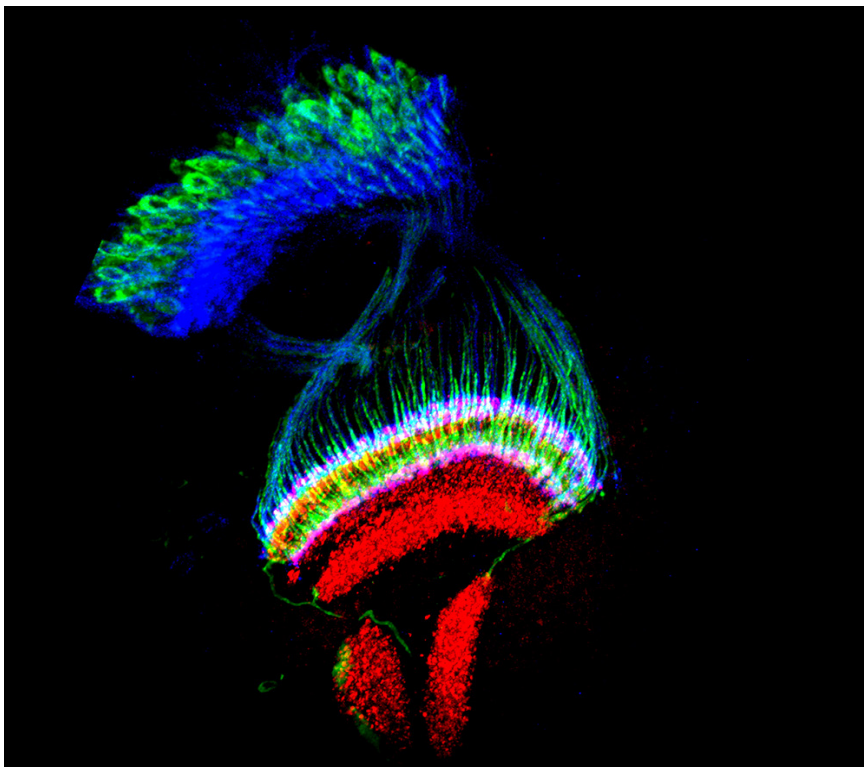
1. Basbaum, A. I., Bautista, D. M., Scherrer, G., & Julius, D. (2009). Cellular and Molecular Mechanisms of Pain. *Cell*, 139(2), 267–284. <http://doi.org/10.1016/j.cell.2009.09.028>.
2. Bautista, D.M., Wilson S.R., & Hoon, M.A. (2014). Why we scratch an itch: the molecules, cells, and circuits of itch. *Nature Neuroscience*, 17, 175–182. doi:10.1038/nn.3619.
3. Cameron, P., Hiroi, M., Ngai, J.,

- & Scott, K. (2010). The molecular basis for water taste in *Drosophila*. *Nature*, 465(7294), 91–95. <http://doi.org/10.1038/nature09011>.
4. Feldman, D.E. (2009). Synaptic Mechanisms for Plasticity in Neocortex. *Annual review of neuroscience*, 32, 33–55. 10.1146/annurev.neuro.051508.135516.
5. Huth, A. G., Lee, T., Nishimoto, S., Bilenko, N. Y., Vu, A. T., & Gallant, J. L. (2016). Decoding the Semantic Content of Natural Movies from Human Brain Activity. *Frontiers in Systems Neuroscience*, 10, 81. <http://doi.org/10.3389/fnsys.2016.00081>.
6. Jadhav, SP, Wolfe J., & Feldman DE. (2009). Sparse temporal coding of elementary tactile features during active whisker sensation. *Nature Neuroscience*, 12, 792–800. doi:10.1038/nn.2328.
7. Kim, H., Kirkhart, C. & Scott, K. (2017). Long-range projection neurons in the taste circuit of *Drosophila*. *eLife*, 6, e23386. <http://doi.org/10.7554/eLife.23386>.
8. Morita, T., Kang, H., Wolfe J., Jadhav, SP, & Feldman DE. (2011). Psychometric Curve and Behavioral Strategies for Whisker-Based Texture Discrimination in Rats. *PLoS One*, 6(6), e20437. <http://doi.org/10.1371/journal.pone.0020437>.
9. Purves, D., Augustine G.J., Fitzpatrick D., Hall W.C., LaMantia A., Mc-Namara, J.O., & White, L.E. (Eds.). (2008). *Neuroscience*. (4th ed.). Sunderland, MA: Sinauer Associates, Inc.
10. Shiozaki, H.M., & Kazama, H. (2017). Parallel encoding of recent visual experience and self-motion during navigation in *Drosophila*. *Nature Neuroscience*, 20, 1395–1403. doi:10.1038/nn.4628.

IMAGE SOURCES

11. J. Luo, C.H. Lee, Eunice Kennedy Shriver National Institute for Child Health and Human Development, NIH. Optic Lobe of *Drosophila* Brain. Retrieved from <https://www.flickr.com/photos/nih-gov/34060441633>.

Figure 2: An enhanced image of a section from the optic lobe of a *Drosophila* fruit fly.¹¹ A neural signal travels from the blue and green region at the top (photoreceptors in the eye that sense the stimulus) to the red region at the bottom (the brain). Neurons interpret a visual stimulus by communicating signals across the nervous system. In the brain, visual sensory information is integrated with motor information to facilitate the fly’s decision-making process.



FUSION: THE PATH TO LIMITLESS ENERGY

BY MATT LUNDY

Fusion power may not be literally limitless, but for all intents and purposes, it is. Nuclear fusion is the process of combining, or fusing, multiple atomic nuclei. This produces absurd amounts of energy; for scale, all of the energy output from our Sun comes from nuclear fusion. Yet, despite our best efforts, we have never achieved a fusion reaction powerful enough to serve widespread use. Because of the slow progress of commercially viable fusion, many have written it off as a pipe dream, but the immense potential behind this energy source has compelled decades of scientific work and continues to attract significant research efforts.

To understand just how powerful nuclear fusion is, consider the Sun. Our Sun is powered by fusion, and uses its most abundant element, hydrogen, as fuel for the process. The Earth's largest sources of readily available hydrogen are our oceans; as hydrogen constitutes two-thirds of water's molecular makeup, fuel derived from Earth's greatest bodies of water would be plentiful. If we could properly utilize fusion on Earth, the current hydrogen in the oceans would be able to power global human energy consumption at its current levels, for two billion years.¹ If we could access such a vast store of power, energy would practically cease to be a finite resource.

These huge amounts of energy are possible because the process of fusion converts mass to energy, and even a little bit of mass is equivalent to a massive amount of energy. When two atomic nuclei combine, the resulting atom has slightly less mass than the sum of the two individual atoms. This difference in mass is released as energy according to Einstein's famous equation, $E=mc^2$, which tells us that energy equals the product of mass and the speed of light (3×10^8 m/s) squared. This means that just one gram of matter—the mass of a paper clip—completely converted into energy could power a 100 watt light bulb for around 30,000 years, which is about five times longer than recorded human history.

Fusion has many benefits beyond its power. In contrast to fission—the reverse nuclear process that powers our modern-day reactors and bombs—fusion is a cleaner and safer alternative. Fusion solves the two major flaws

of fission: radioactive waste and the possibility of meltdown. Any waste created by current methods of fusion has a half-life of only around a hundred years at most.¹ After a relatively short period of quarantine, the waste is completely safe. Additionally, a fusion meltdown is impossible. A fission meltdown, in which the reactor core or shielding overheats and melts, is possible because a fission reactor houses years' worth of energy-producing mass at any one time. It does this because it has to maintain a chain reaction of splitting atoms. On the other hand, a fusion reactor has to be constantly fed mass, meaning that the mass inside the reactor at any given time is tiny, equivalent to that of a few postage stamps. Thus, there is never enough mass for a meltdown to occur.¹

If fusion surpasses fission, then fusion also beats out fossil fuels like coal and gas. It produces no CO₂ or other harmful greenhouse gases. Fusion avoids the other most common fear of fossil fuels too: that one day they will run out. Even if we were one day able to drain every drop of our oceans for energy, hydrogen is the most common element in the universe. Fuel is not a worry for fusion.

For all its immense benefits, however, fusion is simply not currently usable as a power source. As of now, every other energy source discussed is more practical than fusion because those other sources have tangible results. A fusion reactor that can efficiently power any large-scale structure has yet to be built. Performing fusion on Earth essentially means creating a small sun in a lab. This requires immense temperature and pressure to force nuclei together, overcoming the Coulombic force repelling particles of like charge (i.e. the protons in the nuclei.)¹¹ This process takes a lot of energy. The goal of fusion research is to create a reactor that maintains the initial driving energy at levels high enough to sustain continuous fusion. This point is adequately named ignition, analogous to how ignition of a car engine starts a continuous process.

While ignition has not yet been achieved, prom-



“Fusion avoids the other most common fear of fossil fuels too: that one day they will run out.”

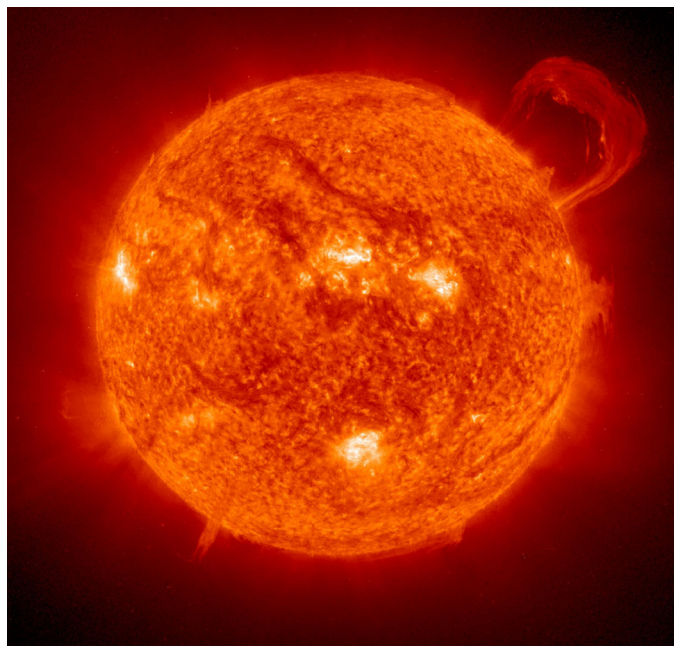


Figure 1: Our Sun is a superb example of the power of nuclear fusion, which serves as the Sun's main energy source.¹²

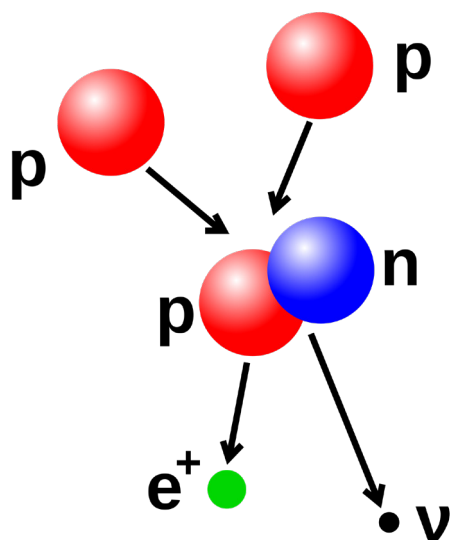


Figure 2: A simple diagram of the physical process of fusion with two hydrogen atoms.¹³

ising progress is being made in the world of fusion research. Recently, the breakeven point was finally reached. In 2013, a team at the US National Ignition Facility achieved an energy gain from “an inertially confined fusion implosion” which surpassed the energy required to start the reaction.¹¹

Although it has taken decades to get to this point, and commercial viability has yet to be achieved, the research and applications for fusion power are only growing. Fully refined power plant designs that would be put into use once ignition is achieved have already been developed, such as the Lawrence Livermore National Laboratory's “Inertial Fusion Power Plant Concept of Operations and Maintenance.”¹⁴ A Fusion Nuclear Science Facility (FNSF) that could be built within the next decade has been designed to facilitate progress towards fusion energy in many ways, such as to obtain information on “reliability, availability, maintainability, and inspectability (RAMI) of fusion nuclear components.”¹⁸

Fusion power has some large scale applications. In fact, the clearest need for fusion power is found in space travel. Long-range space missions require significant leaps in minimizing fuel mass and maximizing energy use. An insignificant amount of lithium (with regards to overall mass) could serve as sufficient fusion fuel. More so, the Fusion Driven Rocket design from NASA makes use of a direct energy-to-propellant fusion system that would not require energy to be transferred into electricity, preventing any loss of energy that would be found in such a step.⁹

While fusion energy is not currently viable, continual progress in research is being made, which reassures the validity of fusion power in the future. As our society pushes further and further technologically, we will need a power source that can handle the exponential growth of energy demands and that can also innovate to allow future fields like space exploration to develop unhindered by energy constraints. With immense benefits in terms of abundance, power, and cleanness, fusion energy is arguably the best candidate for long-term energy needs.

REFERENCES

1. Freidberg, Jeffrey P. *Plasma Physics and Fusion Energy*. Cambridge University Press, 2010.
2. Chapman, John J. “Advanced Fusion Reactors for Space Propulsion and Power Systems.” NASA, NASA, 26 June 2011, <https://tinyurl.com/y76y-hzly>.
3. Reyes, S, et al. “LIFE: a Sustainable Solution for Developing Safe, Clean Fusion Power.” Health Physics., U.S. National Library of Med-

icine, June 2013, www.ncbi.nlm.nih.gov/pubmed/23629070.

4. Anklam, T., Knutson, B., Dunne, A. M., Kasper, J., Sheehan, T., Lang, D., ... Mau, D. (2015). Inertial Fusion Power Plant Concept of Operations and Maintenance. United States. doi:10.2172/1179431.
5. Wendt, Amy, Callis, Richard, Efthimion, Philip, Foster, John, Keane, Christopher, Onsager, Terry, & O'Shea, Patrick. (2015). Applications of Fusion Energy Sciences Research - Scientific Discoveries and New Technologies Beyond Fusion. United States. doi:10.2172/1272148.
6. Beiter, Philipp, & Tian, Tian. (2016). 2015 Renewable Energy Data Book. United States. doi:10.2172/1339347.
7. Destouni, Georgia, and Harry Frank. "Renewable Energy." *Ambio*, Springer Netherlands, 3 June 2010, www.ncbi.nlm.nih.gov/pmc/articles/PMC3357647/.
8. Garofalo, A. M., Abdou, M. A., Canik, John M., Chan, Vincent S., Hyatt, A. W., Hill, D. N., ... Ying, Alice. (2014). A fusion nuclear science facility for a fast-track path to DEMO. *Fusion Engineering and Design*, 89(7-8). doi:10.1016/j.fusengdes.2014.03.055.
9. Slough, John, et al. "Nuclear Propulsion through Direct Conversion of Fusion Energy: The Fusion Driven Rocket." NASA, NASA, 30 Sept. 2012, <https://tinyurl.com/ybgmv56c>.
10. Hayes-Sterbenz, Anna Catherine. (2017). Applications of nuclear physics. *Reports on Progress in Physics*, 80(2). doi:10.1088/1361-6633/80/2/026301.
11. Hurricane, O. A., et al. "Fuel Gain Exceeding Unity in an Inertially Confined Fusion Implosion." *Nature*, Nature, 20 Feb. 2014, www.nature.com/nature/journal/v506/n7488/full/nature13008.html#ref1

IMAGE SOURCES

12. https://upload.wikimedia.org/wikipedia/commons/9/9c/The_Sun_with_Prominence.jpg.
13. https://upload.wikimedia.org/wikipedia/commons/thumb/7/74/Wpdms_physics_proton_proton_chain_1.svg/2000px-Wpdms_physics_proton_proton_chain_1.svg.png.
14. <https://kathrynwarmstrong.files.wordpress.com/2008/10/image.jpg>.



RESTORING THE BROKEN PRAIRIE

BY ANDREANA CHOU

Lush grasses stretch in an endless sea. Wisps of cloud sweep overhead. Bison roam as bees drift from wildflower to wildflower. These are just some of the images evoked by the prairie, a flat and fertile land dominated by grasses. Unfortunately, this vision of the prairie has changed dramatically as agricultural, urban, and livestock developments have taken their toll on the landscape. The North American prairie, commonly known as the Great Plains, encompasses ten U.S. states and parts of Canada. In some parts of the U.S., over ninety-nine percent of prairie has been destroyed.⁴ Luckily, recent understanding of our negative impact on the prairie has generated much discussion over methods of restoration and the future of prairie conservation, as we scramble to save what we have degraded.

Prairies have suffered various damages like species imbalance, fragmentation, soil degradation, and more, due to human encroachment. One of the main disruptors

of the prairie is its biggest industry: agriculture. Factory agriculture, which seeks to maximize profit at the expense of the environment, has grown significantly in order to support the needs of the growing population. Copious use of fertilizer and manure produces excess nitrogen, which distorts the native prairie ecosystem by shifting the balance in soil minerals and affecting soil microbial communities.^{4,6} Microorganisms, including bacteria and fungi, release digestive enzymes that are necessary to maintain soil fertility.¹⁰

Soil microbe imbalance exemplifies how the problems that affect the prairie are not always visible to the naked eye, and demonstrates the importance of restoration methods that manipulate the ecosystem from the ground up. Decreased biodiversity can be attributed to habitat loss and fragmentation.⁹ Fragmentation occurs when a large region is divided into smaller, more isolated communities. Urban developments, fencing, and clearing land for agriculture

have contributed to the fragmentation of the prairie, which causes habitat loss, decreased movements and lack of interaction due to isolation. One such disruption is that of the intricate system of prairie streams, which are able to support up to nearly half of all prairie birds.⁸ Preserving prairie streams is thus crucial to facilitating nutrient transfer, irrigating plants, and attracting animals. Imbalance in biodiversity is also caused by competition between native species and exotic ones. This emphasizes the importance of soil, for exotic plants fare better in nitrogen-rich soil than native plants. Without continuous management or burning, these invasive species build up into monocultures that exclude native prairie species. The consequences of human actions thus make restoration a more challenging job than just reserving spaces and planting a few native species.

The most common methods used to restore the prairie are ones that mimic natural cycles. In the past, prairies have always

“IN SOME PARTS OF THE U.S., OVER NINETY-NINE PERCENT OF PRAIRIE HAS BEEN DESTROYED.”

been “managed.” In fact, fires started by indigenous peoples, to stimulate growth of new grasses and attract animals, were more frequent than lightning-caused fires. In addition, herbivores such as bison would graze to ensure space for new crops of grasses.² Scientists have also attempted to forcibly introduce certain native species. Studies show that plant diversity in tallgrass prairies is proportional to bee diversity, and a strong bee population likewise ensures the growth of future plants.⁷ The diversity and abundance of birds is also indicative of the ecosystem’s health. To encourage bird and bee diversity, conservationists utilize seed mixes that contain native plants, as well as chronosequence methods (which is when a set of sites share similar features but different ages) in or-

der to create a variety of habitats. Management burning is another popular method which maintains stable community composition by clearing debris. The timing of fires is important, as dormant season (spring and fall) burns favor certain species and are therefore less effective than growing season (summer and winter) burns.⁴ The issue of excess nitrogen can be remedied by adding carbon to soil via sucrose and sawdust. The conditions for this to work, however, are very specific: weeds must be nitrophilic or nitrogen-favored and must suppress native species in the absence of carbon. When these conditions are met and enough carbon is added, the decrease in nitrate and weed biomass results in increasing light penetration, soil water content, and biomass of native prairie species.⁵ Other methods, such as altering and building water

channels, utilizing new cattle fences that allow small animals to pass through the bottom, and rooting out weeds, are more mechanical. Given the intricacies of the prairie system, from soil composition to animal diversity, conservationists must consider all factors when rebuilding the prairie.

One major problem is that the current altered environment, known as a degraded system, does not respond to restoration methods the way a natural prairie would. Degraded systems are shaped by lack of landscape connectivity and sufficient seed source,¹ and different recovery patterns in the soil that are influenced by root biomass, root quality, and soil microbes.³ Alternative models are then used to better predict the outcomes of certain actions. An alternative model combines both positive effects of the restoration method and factors affecting the alternative states. The complexity of the prairie means scientists must identify multiple factors that can make a system resilient to management. Conservationists must also prioritize which obstacles need to be addressed first. The main goal is to disturb the constraints that keep the prairie locked in its degraded state, and in order to determine whether their actions have any effect, conservationists must repeatedly check and recalculate the constraints.

When one thinks of saving the environment, the prairie is often pushed to the side in favor of rainforest and ocean habitats. However, the prairie holds just as much cultural significance, biodiversity, and economic value as other ecosystems, which makes rebuilding it more important than ever. Soil degradation, lack of biodiversity, and rigidity of alternative states are just a few of the problems that prairies are currently facing. In order to remedy what humans have inflicted upon the prairie, conservationists lean towards restoration efforts that are as ‘natural’ as possible. Although some methods may seem futile, it is important to keep in mind that prairies, and ecosystems in general, require patience and innovation when breathing them back to life.

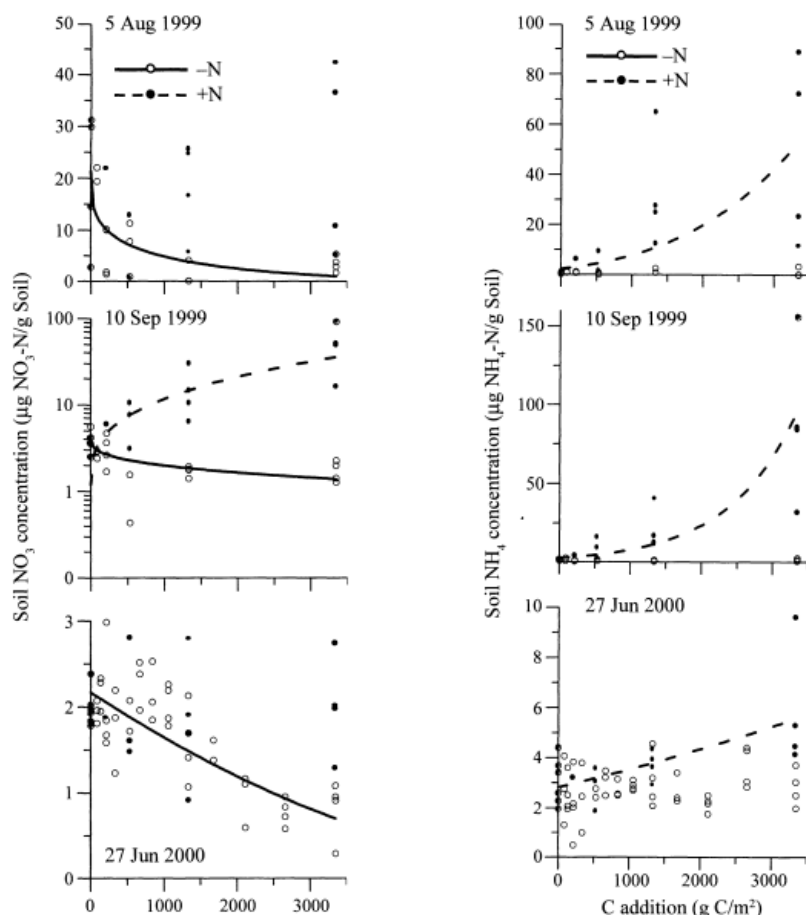


Figure 1: Soil NO₃ and NH₄ concentration regressed with carbon addition.¹²

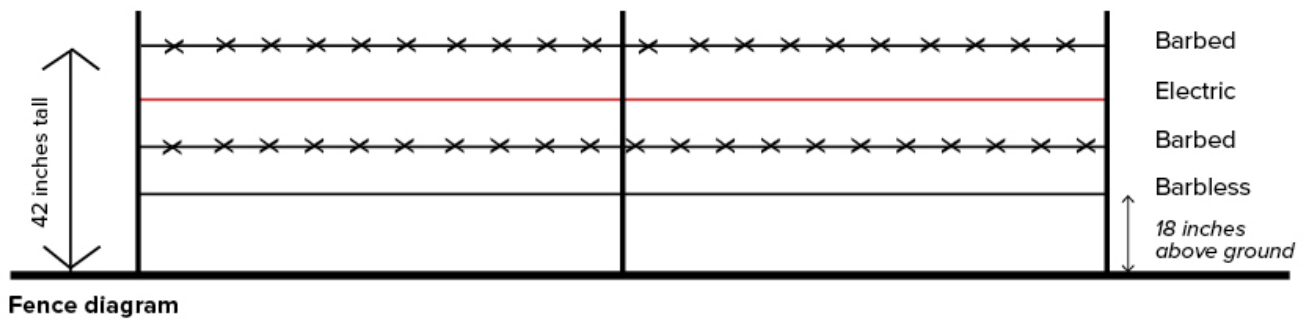


Figure 2: A fence that is barbless on the bottom increases connectivity and makes movement of animals easier.¹³

“Given the intricacies of the prairie system, from the soil composition to animal diversity, conservationists must consider all factors when rebuilding the prairie.”

REFERENCES

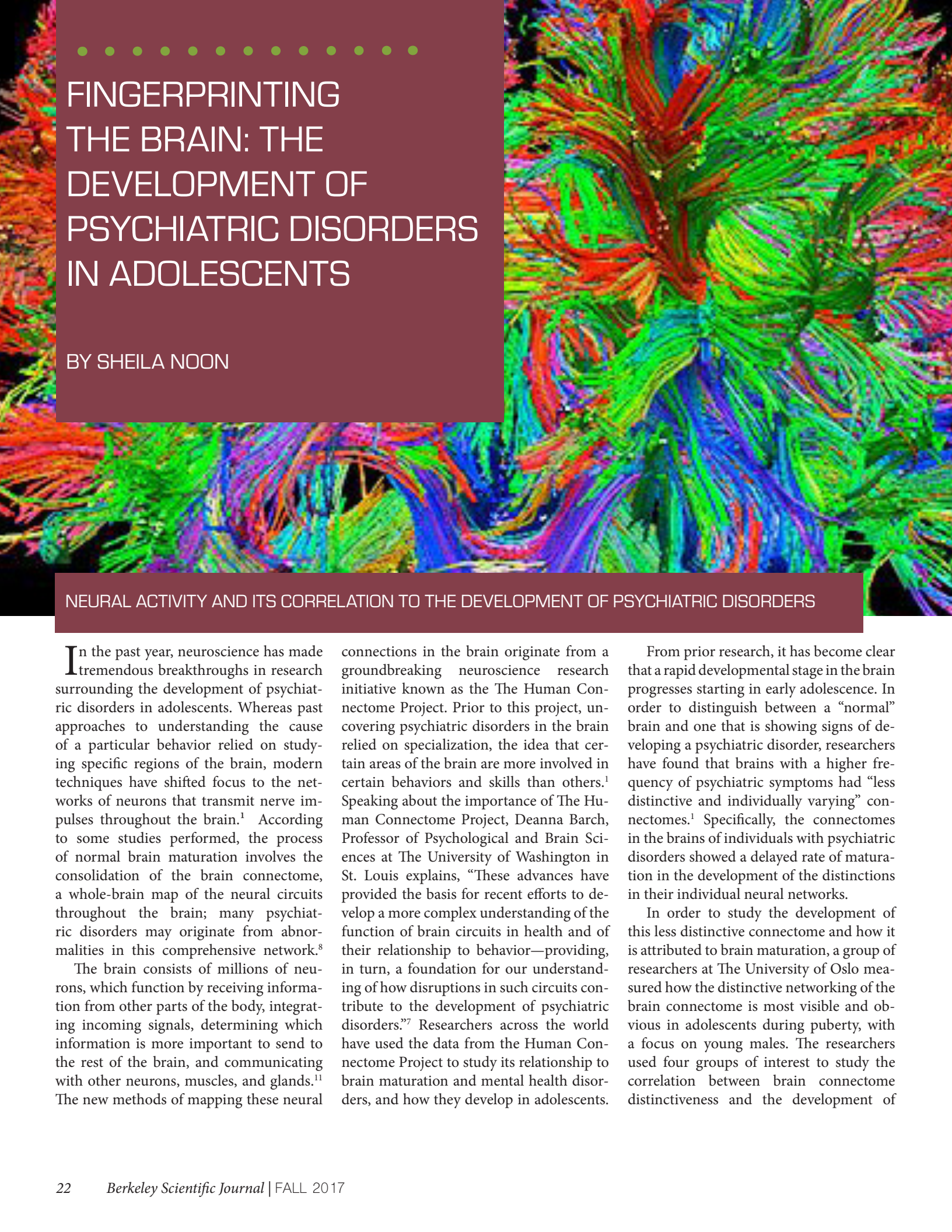
1. Lynch, B. M. (23 August 2017). Long-term study aims to understand prairie ecology after farmland is forsaken. The University of Kansas.
2. Helzer, C. (5 June 2011). The Myth of Self-Sustaining Prairies. The Nature Conservancy - The Prairie Ecologist.
3. Suding, K. N., Gross, K. L., Houseman, G.R. (2004). Alternative states and positive feedback in restoration ecology. Trends in Ecology and Evolution, 19 (1), 46-53.
4. Van Dyke, F., Van Kley, S. E., Page, C. E., Van Beek, J. G. (2004). Restoration Efforts for Plant and Bird Communities in Tallgrass Prairies Using Prescribed Burning and Mowing. The Journal of the Society for Ecological Restoration, 12 (4), 575-585.
5. Blumenthal, D. M., Jordan, N. R., Russelle, M. P. (2003). Soil carbon addition controls weeds and facilitates prairie restoration. Ecological Applica-

tions, 13 (3), 605-615.

6. Baer, S. G., Bach, E. M., Meyer, C. K., Du Preez, C. C., Six, J. (2015). Below-ground Ecosystem Recovery During Grassland Restoration: South African Highveld Compared to US Tallgrass Prairie. Ecosystems, 18 (3), 390-403.
7. Harmon-Threatt, A., Chin, K. (2016). Common Methods for Tallgrass Prairie Restoration and Their Potential Effects on Bee Diversity. Natural Areas Journal, 36 (4), 400-411.
8. Restoring the Prairie. American Prairie Reserve.
9. Wilson, M. C., Chen, X., Corlett, R. T., Didham, R. K., Ding, P., Holt, R. D.,... Yu, M. (2016). Habitat fragmentation and biodiversity conservation: key findings and future challenges. Landscape Ecology, 31 (2), 219-227.
10. Helzer, C. (24 March 2014). A Primer on Soil Microbes -- an Interview with Sarah Greaves. The Nature Conservancy - The Prairie Ecologist.

IMAGE SOURCES

11. <https://www.flickr.com/photos/usfws-mtnprairie/28559617561>
12. <http://www.jstor.org/stable/pdf/4134681.pdf>
13. <https://www.americanprairie.org/project/fences>



FINGERPRINTING THE BRAIN: THE DEVELOPMENT OF PSYCHIATRIC DISORDERS IN ADOLESCENTS

BY SHEILA NOON

NEURAL ACTIVITY AND ITS CORRELATION TO THE DEVELOPMENT OF PSYCHIATRIC DISORDERS

In the past year, neuroscience has made tremendous breakthroughs in research surrounding the development of psychiatric disorders in adolescents. Whereas past approaches to understanding the cause of a particular behavior relied on studying specific regions of the brain, modern techniques have shifted focus to the networks of neurons that transmit nerve impulses throughout the brain.¹ According to some studies performed, the process of normal brain maturation involves the consolidation of the brain connectome, a whole-brain map of the neural circuits throughout the brain; many psychiatric disorders may originate from abnormalities in this comprehensive network.⁸

The brain consists of millions of neurons, which function by receiving information from other parts of the body, integrating incoming signals, determining which information is more important to send to the rest of the brain, and communicating with other neurons, muscles, and glands.¹¹ The new methods of mapping these neural

connections in the brain originate from a groundbreaking neuroscience research initiative known as the The Human Connectome Project. Prior to this project, uncovering psychiatric disorders in the brain relied on specialization, the idea that certain areas of the brain are more involved in certain behaviors and skills than others.¹ Speaking about the importance of The Human Connectome Project, Deanna Barch, Professor of Psychological and Brain Sciences at The University of Washington in St. Louis explains, “These advances have provided the basis for recent efforts to develop a more complex understanding of the function of brain circuits in health and of their relationship to behavior—providing, in turn, a foundation for our understanding of how disruptions in such circuits contribute to the development of psychiatric disorders.”⁷ Researchers across the world have used the data from the Human Connectome Project to study its relationship to brain maturation and mental health disorders, and how they develop in adolescents.

From prior research, it has become clear that a rapid developmental stage in the brain progresses starting in early adolescence. In order to distinguish between a “normal” brain and one that is showing signs of developing a psychiatric disorder, researchers have found that brains with a higher frequency of psychiatric symptoms had “less distinctive and individually varying” connectomes.¹ Specifically, the connectomes in the brains of individuals with psychiatric disorders showed a delayed rate of maturation in the development of the distinctions in their individual neural networks.

In order to study the development of this less distinctive connectome and how it is attributed to brain maturation, a group of researchers at The University of Oslo measured how the distinctive networking of the brain connectome is most visible and obvious in adolescents during puberty, with a focus on young males. The researchers used four groups of interest to study the correlation between brain connectome distinctiveness and the development of

psychiatric disorders. The brain activity of the participants in this study were examined using Functional Magnetic Resonance Imaging (fMRI), a tool that detects changes in blood flow in the brain relative to its concentration of oxygen. Based on clinical symptom scores, these four groups included a healthy control group, a group exhibiting symptoms of attention deficit disorder, a group showing initial symptoms of schizophrenia, and a group showing initial symptoms of depression. Their findings showed that the groups with clinical symptoms of initial signs of schizophrenia and depression had a delayed rate of maturation in the brain. As these psychiatric disorders progressed, the data showed that connectome differences did not appear during the ages of 8-14, showing that these distinctions develop during puberty.² Tobias Kaufmann, one of the lead researchers on the team, concluded from their data that the brains of adolescents with psychiatric symptoms contained a delay in the individualization of their brain connectome. In other words, the presence of less distinct neural circuits in the brain was correlated with a higher probability of the existence of symptoms of mental illness.³

In order to explain how these psychiatric disorders develop in adolescents, it is necessary to understand the concept of neuroplasticity, the brain's ability to reorganize itself by strengthening or weakening neural connections in response to development and methods of use. Another group of researchers reached this conclusion by focusing on symptoms of depression in adolescents, and how it can preface depression in adulthood. They created four sub-groups out of 243 adolescents, including a control group with low levels of depression, two groups with a decrease in symptoms, and one with an increase in symptoms. They tested the four depression trajectory groups on both whole brain and regional levels, finding that the groups varied in mean differences in nodal efficiency, a measure of how efficiently the network of neurons exchanges information in regards to the level of individual connections between regions in the brain. The groups with symptoms of depression showed a deviation in the normal development of network consolidation in the brain, suggesting that this might be an indicator of an individual's

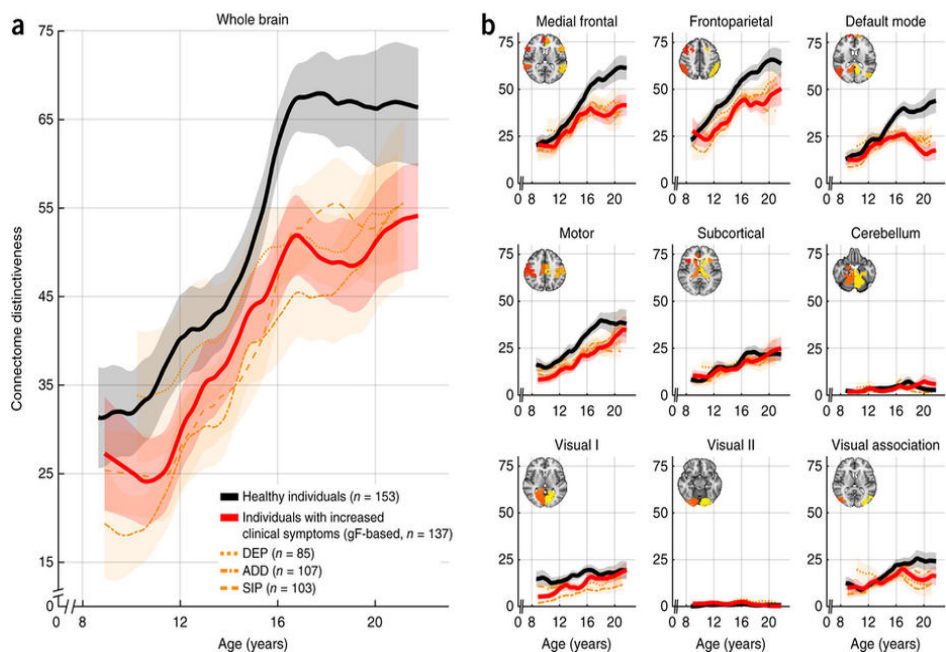


Figure 1: These graphs represent (a) analysis of whole-brain connectivity and (b) analysis of nine sub-networks of the brain (medial frontal, frontoparietal, default mode, motor, subcortical, cerebellum, visual I, visual II, and visual association). Individuals with a higher clinical symptoms score showed a lack of brain maturation in terms of brain connectome distinctiveness.¹³

predisposition to depression in the future.⁹

Since just the beginning of this year, neuroscientists have already made great strides in understanding the maturation of our individual neural networking in the brain and its subsequent correlation in the development of psychiatric disorders.¹ Advancements in the understanding the individuality of our distinct connectomes and the delayed rate of maturation in the brain contributes to the viability of recognition and treatment of these abnormalities, propelling this branch of neuroscience research into the future.

REFERENCES

1. Galvan, Adriana. Adolescence, Brain Maturation and Mental Health. *Nature Neuroscience*, March 2017.
2. Kaufmann, Tobias; Alnæs, Dag; Doan, Nhat Trung; Lycke Brandt, Christine; Andreassen, Ole; Westlye, Lars. Delayed Stabilization and Individualization in Connectome Development are Related to Psychiatric Disorders. *Nature Neuroscience*, January 2017.
3. Fingerprinting Young Brains: New

"Connectomes resemble a fingerprint; they are distinct from person to person and evolve during adolescence."

“The presence of less distinct neural circuits in the brain was correlated with a higher appearance of symptoms of mental illness.”

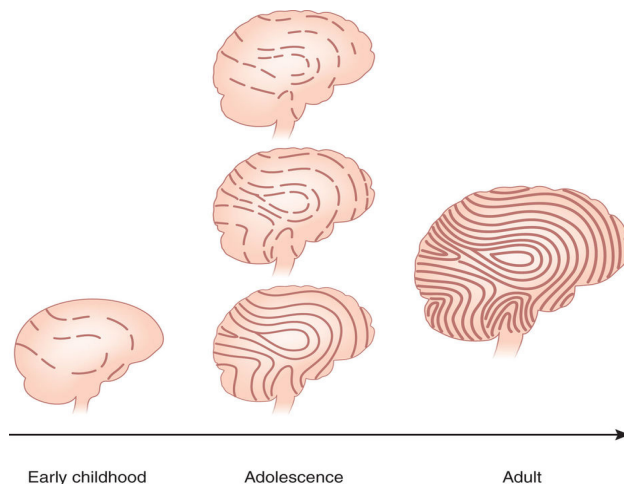


IMAGE SOURCES

12. <https://i.pinimg.com/736x/99/eb/8d/99eb8d33ca915c748a2d3ee8b-feaf326--psychedelic-art-physiology.jpg>.
13. http://www.nature.com/neuro/journal/v20/n4/fig_tab/nn.4511_F3.html.
14. <http://www.nature.com/neuro/journal/v20/n4/full/nn.4530.html?foxtrotcallback=true>.

Figure 2: The brain connectome ‘fingerprint’ transforms into a distinct and individualized network of neural connections during early adolescence. In this study, people with higher levels of clinical symptoms of psychiatric disorders had visibly less distinct connectomes, as seen in the top figure of the brain during adolescence. People whose neural networks revealed greater complexion and individualization had, on average, a lower

Leads for Mental Disorders. Science Daily, February 2017.

4. Cocchi, Luca; Harding, Ian; Lord, Anton; Pantelis, Christos; Yucel, Murat; Zalesky, Andrew. Disruption of Structure-Function Coupling in the Schizophrenia Connectome. *Neuroimage: Clinical*, May 2014.
5. Thijssen, Sandra; Kiehl, Kent. Functional Connectivity in Incarcerated Male Adolescents with Psychopathic Traits. *Psychiatry Research: Neuroimaging*, July 2017.
6. Wilkins; Williams, Lippincott. New Findings on Brain Functional Connectivity May Lend Insights Into Mental Disorders. *Wolters Kluwer Health*, August 2017.
7. Barch, Deanna. Resting-State Functional Connectivity in the Human Connectome Project. *Harvard Review of Psychiatry*, 2017.
8. Whitaker, Kristie; Vertes, Petra; Romero-Garcia, Rafael; Vasa, Fran-tisek; Moutoussis, Michael; Prabhu, Gita; Weiskopf, Nikolaus; Callaghan, Martina; Wagstyl, Konrad. Adolescence is Associated with Genomically Patterned Consolidation of the Hubs of the Human Brain Connectome. *Proceedings of the National Academy of Sciences of the United States of America*, February 2016.
9. Ellis, Rachel; Seal, Marc; Adamson, Christopher; Beare, Richard; Simmons, Julian; Whittle, Sarah; Allen, Nicholas. Brain Connectivity Networks and Longitudinal Trajectories of Depression Symptoms in Adolescence. *Psychiatric Research: Neuroimaging*, February 2017.
10. Essen, David; Smith, Stephen; Barch, Deanna; Behrens, Timothy; Yacoub, Essa; Ugurbil, Kamil. The WU-Minn Human Connectome Project: An Overview. *NeuroImage*, October 2013.
11. The Neuron. Society for Neuroscience, April 2012.

A METEORIC JOURNEY THROUGH THE YEARS

Interview with Professor Alex Filippenko

BY MELANIE RUSSO

Alex Filippenko is a Professor of Astronomy at the University of California, Berkeley. He has been a part of both the Supernova Cosmology Project and the High-Z Supernova Search team which discovered the acceleration of the universe and the possibility of existence of dark energy. Professor Filippenko has received multiple awards for his undergraduate teaching and has been frequently featured in the media. In this interview, we discuss his journey from a young dreamer to a man who has helped revolutionize what we know about physics and the universe.



Professor Alex Filippenko

BSJ: Can you tell us about your journey from your childhood to the remarkable astrophysicist that you are today?

AF: Well, I will give you the decently short version. I was a science nut from my earliest days. I would play with magnets in the first grade and marvel at the little pieces of iron that would stick to the magnet. I was very interested in being able to control magnetism, and someday I hoped to understand it. In freshman year of high school, I requested and received a small telescope for Christmas. So I took it out that night, looked at the stars, and saw one that was brighter than the rest. When I looked into books and realized that the star was Saturn, it knocked my socks off. It didn't matter that millions of people had seen it before me—astronomy became a growing interest. However, I entered college as a chemistry major, because that interested me more at the time. I took an introductory astronomy course and learned that

the large-scale behavior of the universe is dictated by the small-scale interaction of particles. So, I could satisfy all my interests by studying astrophysics. Also, as a chemistry major, you had to take an organic chemistry lab course, and back then, fume-hoods weren't so good. I would pass the organic chemistry lab on my way to another class, notice the smoke coming out of the classroom, and think to myself, "Do I really want to take this class? No." As you can see, it was really an act of self-preservation that I switched majors, because as a chemistry major I would have to deal with explosives. I eventually went on to study astrophysics at Caltech, and then came to Berkeley as a post-doc and stayed on as a faculty member. As an astrophysicist here, I was continuing a project with my advisor from Caltech, Wallace Sargent. We were looking at a bunch of galaxies for evidence of a supermassive black hole. Once I stumbled across an exploding star, and that turned me on to another area of research—supernovae. And I became an expert on that. And then I was invited to participate in a

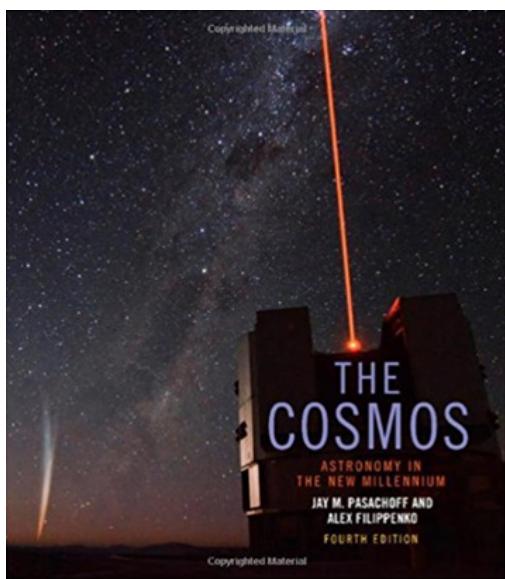


Figure 1: Left: Cover of introductory textbook *The Cosmos: Astronomy in New Millennium*, written by Professor Filippenko and Jay Pasachoff. Right: The High-Z Supernova Search Team receiving the Nobel Prize in Physics in 2011.

project that used supernovae to study the expansion of the universe. I was a spectroscopist and I had access to the greatest telescopes, so I was a natural fit for these research groups. And that, in short, is a summary of how I got here.

BSJ: You worked with two teams to find evidence for the accelerated expansion of the universe. Tell us about your work with these teams.

AF: The expansion of the universe was first proven by Edwin Hubble in the late 1900s. Everyone expected that expansion would be slowing down, because everything pulls on everything gravitationally. The question was: would it slow down so much that it would stop, and re-collapse in on itself, or would it slow down only gradually—like an apple thrown at a velocity equal to Earth's escape speed—in which case it slows down but still eternally expands? Those were the theories we

were working with. But, in fact, it was neither of these things, and, actually, in the last 5 billion years the expansion has been speeding up. So this is what we discovered.

BSJ: Data shows that when we look out into the stars, everything is moving away from us. Are we some sort of a center point, or is the expansion relative?

AF: When we saw this, we thought, do these galaxies not like us? Do we smell? Is it something we said? Are we from Stanford? (*laughs*) Or are all these galaxies lactose-intolerant because they are all moving away from the Milky Way? Get it? (*laughs*) But no, we are not some sort of a central position. Picture a bunch of ping-pong balls on a rubber hose. If you stretch the hose, you will see that no matter which ping-pong ball I'm on I would see the others move away. So, we are not special.

BSJ: What is like to be an astrophysicist? What's a day in your life?

AF: Oh, it's fantastic. You get to explore the workings of the universe, pose questions that are of genuine intellectual interest, and try to come up with ways to answer them. And in your own way, you're always trying to contribute to the advancement of knowledge in science. You can't expect to make a gigantic discovery, but occasionally that happens!

BSJ: You're the expert on science popularization. How many TV shows/documentaries have you been on?

"Once I stumbled across an exploding star, and that turned me on to another area of research—supernovae."

AF: I think about a thousand. Including weather channels. They ask me about the weather patterns on other planets and want me to explain what it would be like if such weather patterns were observed on Earth.

BSJ: What are your top qualities that make you so good at what you do?

AF: Well, I guess I'm very enthusiastic. I wear my passion on my sleeves. Students react well to that. There is nothing worse than a lecturer who may be very strong in their field but presents information in a dry way. Preparation and clarity are also very important. I deliberately attempt to realize that what I'm presenting is difficult and spend time thinking of analogies and ways to make the material more comprehensible. I also make time for my students—like yesterday, when I held a two-hour bull-session on quantum physics which ended up being three-and-a-half hours. And the week of finals, I'll be doing another one. Even though that's 10 extra hours that I'm not being paid for. I'm happy to do it. I also add little bits to my lectures—I wear those weird T-shirts (yes, it's a gimmick, but some students get excited to see what T-shirt I'll be wearing next). I also tell a lot of jokes, and update students on current astronomy events.

BSJ: Do you believe in a god?



Figure 2: Professor Filippenko wearing one of his well-known science T-shirts.

AF: My god is the laws of physics, or whatever it is that produced the laws of physics. I don't think there is anything particular that is guiding us or the laws of physics. And no, I don't believe that there is any sort of spirit overruling us, but I also don't think that any type of science could ever prove or disprove that hypothesis. I think we are the result of 13 billion years of evolution, and yes, it could be extremely rare that this sort of thing occurred, but in an infinite universe, it will have occurred many times. In a finite universe, maybe we're just the only ones. But in an infinite universe, everything that can happen will happen, many times. We still can't answer why there is something rather than nothing, and I'd rather allow religion and science to coexist and agree to be friends. When it comes to religions, I respect all the good they've done, but I have a problem when they start to enforce their beliefs on others.

BSJ: Do you believe that there is a multiverse?

AF: Well, once you've produced one universe, it's hard to see why it would be limited to one. And if it's not limited to one universe, why would there be three, or eight? In other words, what limits our universe? Nothing. The only way to detect this would be to observe two universes collide, but we don't have such technology yet.

BSJ: What advice do you have to aspiring scientists?

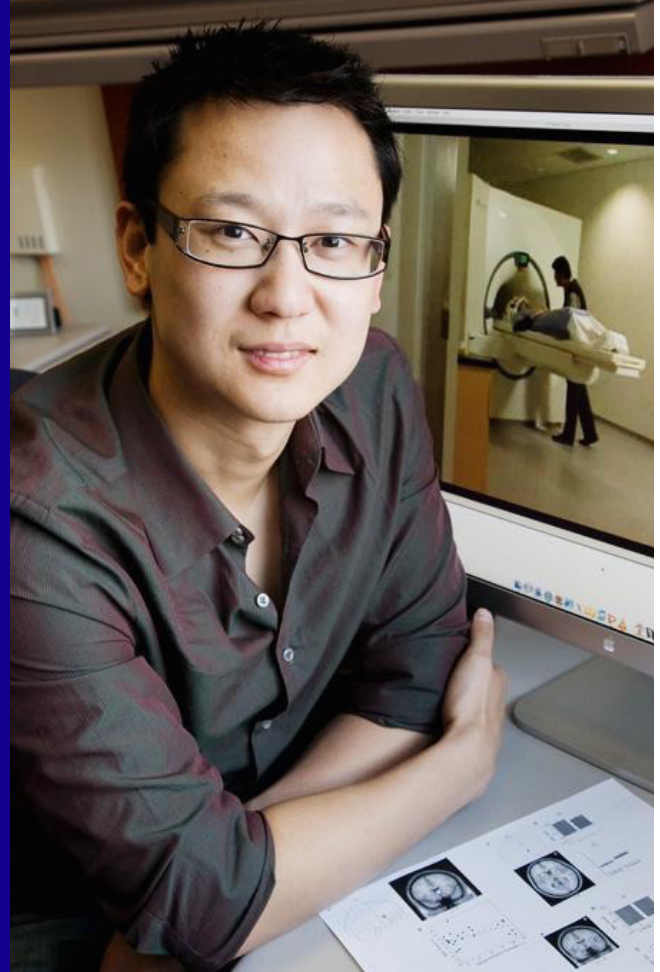
AF: Study hard, work hard, be enthusiastic, and take advantage of opportunities that fall in your lap. Most importantly, don't be hard-going when things don't go as planned. Research is not a linear process. Don't be disappointed if in your own view you're only making incremental advancements. That's what most scientists do, and it's only the lucky few that have the brilliant breakthroughs à la Newtons and Einsteins. Don't compare yourself with Newton and Einstein—they lived in a different plane of existence. Set your sights high, but be satisfied with whatever steps you are taking.

REFERENCES

1. <http://astro.berkeley.edu/faculty-profile/al-ex-filippenko>
2. <https://i.pinimg.com/originals/a6/12/c1/a612c1156cb1a11658b517962f85609c.jpg>

NEUROMARKETING: USING FMRI TO STUDY THE BRAINS OF CONSUMERS

BY KIM DO, YINUO HAN, CASSIDY
HARDIN, YANA PETRI, AISWARYA
SANKAR



Ming Hsu is an Associate Professor at the Haas School of Business and Helen Wills Neuroscience Institute at the University of California, Berkeley. He received his Ph.D. in Economics from the California Institute of Technology and now heads a neuroeconomics lab at Berkeley that studies consumer choice and social behavior. In this interview, we discuss how fMRI can be used to study the consumer brain and brand personality without resorting to biased self-reports.

BSJ: Given your background in Political Science and Economics, how did you get into the field of neuromarketing?

MH: I think the common thread is the need for all of these fields to understand human decision-making. What distinguishes my work and that of those like me is that we also look into the biological mechanisms in addition to behavior and societies. How I got into the field is a bit more circuitous. As you mentioned, I studied Economics and Political Science as an undergrad at the University of Ar-

izona, but at the same time I was working in a cognitive neuroscience lab. That was when folks in the social sciences first started working with functional magnetic resonance imaging (fMRI). It was around that time that I saw a fascinating presentation from a group of economists who used fMRI to study how people made economic decisions. I thought that this was one of the coolest things that I had ever seen and soon started working with them. I had no idea at the time that these were economists of some renown, especially Vernon Smith, the director of the lab who won the 2002 Economics Nobel Prize. The second major influence was in my PhD studies working with my thesis advisor Colin Camerer. He has this sixth sense for knowing what is going to be the next generation of cutting-edge research questions. For example, before Caltech even had an fMRI scanner, he would tell me to design experiments applying brain imaging to economic questions, because he thought this was going to be a hugely important set of questions. Turns out he was completely right.

BSJ: What kind of information can neuromarketing extract from data that could be of interest to people in the field of marketing?

MH: The field is moving incredibly fast, so it's hard to say something that isn't at risk of becoming obsolete in the near future. But in general, I would say that we can use neuroscience to address some of the long-standing skepticism that people have toward self-report measures, like focus groups and surveys. We can use neuroscience to validate whether what consumers are telling us reflects what is going on in their brains. In the same way that people don't necessarily trust interviews, there is a lot of skepticism surrounding focus groups, so not everyone will report their true thoughts. We can use neuromarketing to avoid bias in consumers' answers and confirm that the information we derive from consumers' brains is consistent with their self-reports. We can also use neuroscientific tools to increase the precision of

our measures; for example, consumer engagement. Finally, we can use neural measures to forecast how consumers will react to a branding or an advertising campaign. Can we scan a brain and figure out signatures that are going to allow us to project what's going to happen after the product is launched?

BSJ: In your research you talk about several methods that you use for brain response imaging. Can you go over briefly what these methods are and why you chose to use fMRI specifically?

MH: There are three primary imaging methods used in human neuroscience: fMRI, electroencephalography (EEG), and positron emission tomography (PET). PET is largely reserved for medical imaging because it involves injection of radioactive tracers. So for neuromarketing we are typically talking about fMRI and EEG only. In terms of how we choose a specific method, the primary tradeoff is between cost and portability on the one hand, and the anatomical specificity on the other. If you care about portability and are on a limited budget, EEG is the likely choice because fMRI requires a 3-ton machine that costs about \$1 million, so good luck wheeling it into a movie theater, for example. The downside of EEG is that it doesn't have very good spatial resolution, so you can't easily tell which brain



Figure 1: Difference between (a) the range of consumer choice in the laboratory and (b) the range of consumer choice in the real world.¹

regions are contributing to your signal. So if you care about where things are happening in the brain, fMRI is the better tool.

BSJ: In one study,³ you used machine learning and fMRI to study brand personality—human-like characteristics that consumers associate with brands, like excitement, competence, sincerity, etc. Could you explain for our readers what this brand personality framework is and how you mapped neural activity in certain regions of the brain to specific brand personalities?

MH: Brand personality is the idea that people can think about brands in anthropomorphic terms. For example, people rate Campbell's Soup as high in terms of being "wholesome," or Google as being "imaginative." It is intuitive but sometimes

controversial because, beyond self-reports, there are few ways to actually validate what consumers think about a brand. For example, a skeptic can argue that people don't spontaneously think of these attributes without being explicitly prompted by the questionnaire. What's missing is the possibility for objective independent verification, very much like DNA evidence in forensic analysis. Our answer to this is to put people in an fMRI scanner. In the study you mentioned, we asked them to think about well-known brands like Gucci, Apple, Google, or Ford. After the experiment, we asked them to take the brand personality survey. We then showed that we could use their brain activity to predict how they would describe brands in the subsequent survey. Because our participants didn't know about the survey when they were being scanned, there was no risk that questions could bias their thoughts during the scanning session. This method is not perfect, but it provides some of the first indications that we can use the brain to validate consumer self-reports, much like DNA evidence can be used to validate the account of a witness.

BSJ: In the same study, how were neural responses to brands extracted and then mapped to one of the five personality features?

MH: This is where we used ideas from machine learning. We used 42 out of 44 brands as a training set to develop a predictive function for brand personality based on a pattern of brain activity. We learned how consumers respond to different brands and tried to explain this re-

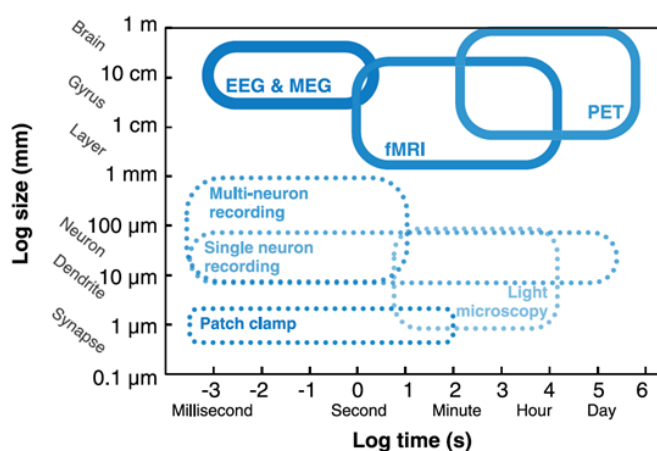


Figure 2: The technology used to study the consumer brain. While EEG is very popular because it is not very expensive, fMRI offers higher spatial precision.²

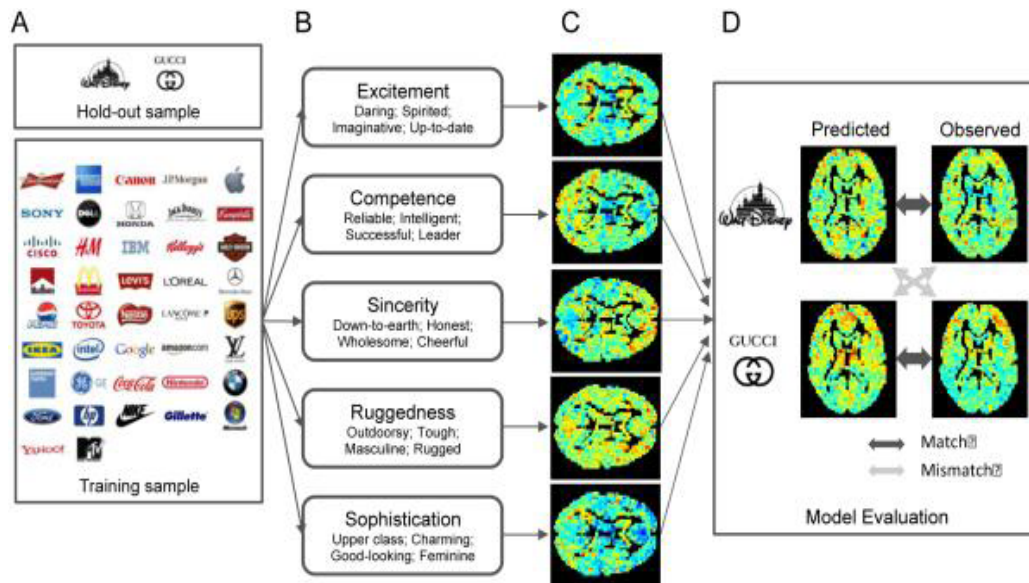


Figure 3: This figure from Ming Hsu's study³ maps how the scientists use hold-out and training brands to create a brain map of the five personality features from which consumer responses to unknown brands can be predicted.

sponse in terms of five personality traits: excitement, competence, sincerity, ruggedness, and sophistication. We used two “hold-out” brands to check whether we could correctly predict brand personality based on the information derived from the training sample.

BSJ: What are the ethical implications of studying the brains of consumers with fMRI?

MH: Let me give an example first involving a collaboration of ours with a sports team. There, one question that we could have asked was, “How can we use the brain to figure out what prices people are willing to pay for tickets?” Or, “How can we figure out what people would like to pay, and what price would extract the maximum amount of profit?” That would have been a really terrible study. The fans would’ve hated it. The team would hate it because they, hopefully, care about what the fans think. Instead, we asked, “How can we use neuroscience to help the team figure out how to deliver a better fan experience?” That is a question that can benefit everyone! More generally, I think this is a challenge facing every new technology, from neuromar-

keting to genetics to artificial intelligence. They can be used with destructive consequences, but also for positive ends. These technologies are simply tools, and the way we use them often determines whether they have destructive or positive consequences.

BSJ: If you were giving advice to a student who wants to do research in the future, what would you say?

MH: Get involved early, whether as a research assistant in a lab or an internship in a company! Start doing what you are passionate about. Also, try to select an area where you have a comparative advantage. For example, when I started neuromarketing, this area of research was new. I was able to pick up the material even faster than my advisors because I had more time. There are so many exciting things that are happening! I’ve had students start creative new research projects because they follow the most recent developments in AI, social media, and other technological advancements.

REFERENCES

1. Hsu, M. (2017). Neuromarketing: Inside the mind of the consumer. *California Management Review*, 59(4), 5-22.
2. Hsu, M., & Yoon, C. (2015). The neuroscience of consumer choice. *Current opinion in behavioral sciences*, 5, 116-121.
3. Chen, Y. P., Nelson, L. D., & Hsu, M. (2015). From “where” to “what”: distributed representations of brand associations in the human brain. *Journal of Marketing Research*, 52(4), 453-466.



UNDERSTANDING NEGATIVE SYMPTOMS IN SCHIZOPHRENIA: A HOLISTIC APPROACH



Interview with Professor Ann Kring

BY NIKHIL CHARI, KIM DO, YINUO HAN, CASSIDY HARDIN, WHITNEY LI, ELENA SLOBODYANYUK

Dr. Ann Kring is a Professor of Psychology at the University of California, Berkeley. Professor Kring's research group focuses on understanding emotional and cognitive processes in psychopathology. In this interview, we discuss Professor Kring's study of negative symptoms and social decision-making in individuals with schizophrenia.

BSJ: How did you get involved in the field of psychology and the study of emotional processing in schizophrenia?

AK: I first got involved in psychology when I was in high school—I took an introductory psychology class and found it fascinating. So I decided that I wanted to major in psychology before I even went to college. Once I got to college, I got involved in research on visual perception, and I was learning how to program computers. I wanted to continue either that or neuropsychology research in graduate school, as I became interested in brain-behavior relationships. After applying to graduate schools, I went to the State University of New York at Stony Brook to study neuropsychology. When I got there, it turned out there wasn't any neuropsychology research going on. So I was scrambling around, trying to figure out what I was going to do as a graduate student. There was a faculty member who was doing research in schizophrenia, and I figured I would



Professor Ann Kring

be able to apply neuropsychology to that. I started learning about schizophrenia and quickly became fascinated by it. All of the myths I had about it were quickly disproven. It was also during that time that I started to read about emotion. I combined schizophrenia and emotion in my dissertation study, and I've been doing the same thing ever since. It wasn't what I entered graduate school for, or what I thought I would end up doing. But once I got to graduate school and started learning about something that I didn't know anything about, it fascinated me and really launched my entire career to read about emotion. I combined schizophrenia and emotion in my dissertation

study.

BSJ: What are negative symptoms, and what specific negative symptoms are found in people with schizophrenia?

AK: Within schizophrenia there are positive and negative symptoms. Positive symptoms refer to having an excess of things that people do not ordinarily have, like hearing voices or having delusions. Negative symptoms refer to not having things that people typically do. There are five different negative symptoms. *Flat affect* is a symptom where people do not have outward expressions of emotions. Individuals may not change their face at all, not laugh, not frown, speak in a very monotone voice, or avoid any eye contact. It creates tremendous social consequences for someone who has that symptom. Another symptom is *anhedonia*, which means “without pleasure.” People with this symptom do not experience enjoyment in things from daily life. *Avolition* is a symptom where people have a hard time motivating themselves to do things. *Asociality* is a symptom of lacking social interactions. Here, people prefer to be alone, and they find being around other people confusing and anxiety-provoking. The last negative symptom is called *alogia*, “without speech.” It’s not that people with schizophrenia are mute, but it means that they do not say very much and will not elaborate on their thoughts. The ones that I focus on the most within my research are the lack of outward expression and anhedonia. We are just now beginning to study asociality and avolition; we are slowly making our way through the negative symptoms.

BSJ: What is the Clinical Assessment Interview for Negative Symptoms (CAINS), and how do you implement it in your studies?

AK: The CAINS is a clinical interview designed to assess the five negative symptoms. Part of the problem with older clinical interviews was that they did not cover the entire domain of negative symptoms. They had not kept up with any recent research in the field, particularly for anhedonia, so we added that topic to the CAINS. Clinical interviews are used a lot in research in schizophrenia to get an assessment of how severe someone’s symptoms are. They can also be used in a clinical setting. If someone has a relative with schizophrenia and they take them to see a psychologist or psychiatrist, they can use these interviews to get a sense of the kinds of symptoms they’re experiencing and try to target the treatment toward the most problematic areas for that person. So the goal of

the CAINS is to come up with a better and more comprehensive way to assess all five negative symptoms.

BSJ: Are there any limitations that remain in the CAINS?

AK: Some people wish it didn’t take as long to administer. The entire test can take 20 to 30 minutes to get through all the questions. People always want to be quick: “How stressful is your life? Tell me on a scale of one to 100. Ok, we’re done.” Boom. Spending 20 minutes going through all the questions is a downside in some people’s eyes. However, we think the tradeoff is worth it because we’re really interested in what a person with schizophrenia tells us. A lot of the time, clinical interviews tend to prioritize a mental health professional’s impression over what a person with schizophrenia says. A psychiatrist might talk to somebody for five minutes and make a judgment based on that conversation. In developing the CAINS, we didn’t think that’s a very good approach for a couple of reasons. First, it diminishes the experience of the person with the illness. You have to listen to somebody to really understand what they’re experiencing. We also think that listening to that experience is what’s going to tell us about the symptom, not a mental health professional’s quick assessment. The reason the CAINS takes longer is because we ask a lot of questions, and we’re trying to get the person with schizophrenia to tell us a lot of information about their own experience.

BSJ: In one study,¹ you investigated anticipatory pleasure deficits in people with schizophrenia. What is anticipatory pleasure and how is it characterized in people with schizophrenia?

AK: The symptom of anhedonia prevents people with schizophrenia from experiencing pleasure. In our studies, we show people with schizophrenia a funny film clip or give them a piece of chocolate, and they tell us that they enjoy it just as much as any other person. They don’t seem to exhibit anhedonia. That made us wonder if anhedonia is more complicated than we think. Our emotional experiences really unfold across time. There are two pieces to anticipatory pleasure: predicting how good you’ll feel about something in the future, and how good you feel in the moment based on knowing something positive is going to happen. Anticipatory pleasure, then, is what motivates us to do lots of things. What we’ve learned in our studies is that people with schizophrenia don’t have problems in experiencing in-the-moment pleasure at all, but rather have trouble with anticipatory

“People with schizophrenia don’t have problems in experiencing in-the-moment pleasure at all, but rather have trouble with anticipatory pleasure.”

pleasure, and that might be where the anhedonia lies.

BSJ: Prospections, which are mental representations of the future, are an important component of anticipatory pleasure. You hypothesized that memory tasks would enrich the prospections of people with schizophrenia. How did your results compare to this hypothesis?

AK: Other studies found that people with schizophrenia enjoy chocolate just as much as anybody, but they don’t maintain or savor that response. We hypothesized that people with schizophrenia might have trouble remembering positive events, and that may be why they don’t look forward to things. A lot of the time, we use the past to think about the future; we envision it and then try to make predictions about whether or not we’re going to enjoy it. We conducted a study in which we interviewed people about memories of the past, and people with schizophrenia didn’t seem to differ from people without schizophrenia in the way they were able to remember things that they enjoyed. However, they still had some trouble prospecting, or looking forward. Our hypothesis that memory would be the key to understanding why people with schizophrenia have trouble anticipating that things will be enjoyable wasn’t fully supported. People with schizophrenia can remember positive things, they just don’t necessarily translate that memory into making predictions that things in the future will be enjoyable.

BSJ: How could memory tasks help people with schizophrenia experience greater anticipatory pleasure?

AK: That is a great question! We’re working on building an intervention for people with schizophrenia to help boost their anticipatory pleasure. Part of what we’re doing is trying to boost memory, even though it seems like it’s okay in people with schizophrenia. Without prompting them to think about a memory, they wouldn’t necessarily do it. Most of us automatically think about our past, even if someone doesn’t tell us to. We think about our stored knowledge—it happens almost automatically. But in people with schizophrenia, it doesn’t happen automatically. We found that if we prompt people to explicitly access their memory, that may actually help them uncover stored knowledge or past experiences, and by doing that we may help people think about the future and take pleasure out of it.

BSJ: In another study,² you investigated social outcomes and subsequent decision-making in people with schizophrenia. Why did you choose to study social decision-making?

AK: Since one of the negative symptoms of schizophrenia is asociality, we were interested in trying to understand emotion in a social context. We knew that social interactions are difficult for people with schizophrenia, so we tried to understand what it was about social interactions that they don’t enjoy. Past studies suggested that although people with schizophrenia said that they prefer to be alone, they would report that they were lonely. That’s a big conundrum. So we turned to social interactions to unpack a little about what it is that’s hard about them. You make a myriad of decisions that you don’t even think about in the context of a social interaction. One such decision we focused on was whether or not to trust a person you’re meeting for the first time based on their facial expressions. There’s a lot of literature outside of schizophrenia that shows that we naturally trust someone who is smiling a lot better than someone who is scowling. So we tested to see if people with schizophrenia would benefit from this display.

BSJ: Can you explain the trust game that was implemented in this study?

AK: The trust game was a computer simulation in which participants iteratively interacted with the same person. We had four different characters, and one of them was Bill. You see Bill, and then you have to decide if you want to invest in him. You give him some amount of points—it’s a proxy for money.

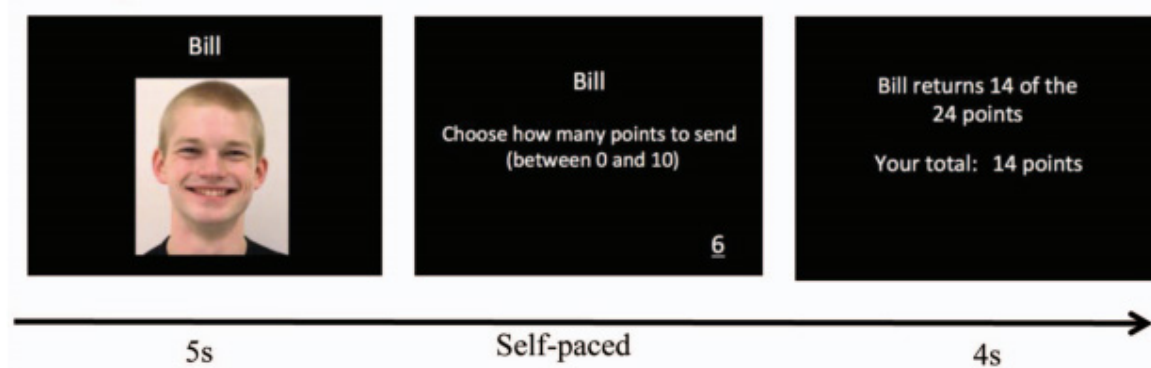


Figure 1: Example of a trust game trial. Participants saw a dynamic clip of a social partner and decided how many points to send to the partner, representing the amount of trust placed in them. Then participants saw the outcome of the interaction by the amount of points returned by the social partner.²

Then Bill will either reward your trust and give you back money, or he will screw you over and take your money away. This kind of game had been done before, and usually people are pretty good at quickly learning who they can and can't trust based on their gains and losses. We did this experiment with two things happening at once: there were the points exchanged, and Bill was either smiling, scowling, or had a neutral display. We found that people with schizophrenia were better than people without schizophrenia at learning when not to trust. That was true whether Bill was scowling or smiling. This could be a reflection of life experiences where people with schizophrenia have trusted others and it wasn't rewarded, or just a slower reaction to benefit from that positive signal. All of that told us something about what interactions may be like for people with schizophrenia. They may be wary to trust, even if it's a friendly person who's smiling and exhibiting trustworthy behavior. That tells us that we need to work with them to help them recognize that they're not ultimately going to get burned by allowing themselves to trust other people.

BSJ: What implications do the results of your research have for developing treatments for people with schizophrenia?

AK: We hope to use the research to develop treatments for a couple of reasons. First, the current frontline form of treatment for people with schizophrenia, medication, doesn't make much of a dent in negative symptoms. Medications can help people stop hearing voices, minimize delusional beliefs, or clear up disorganized thinking, but they don't really help with the lack of outward expression,

the anhedonia, or the lack of motivation. Doing this research, we hope we've drilled in on not just broad negative symptoms, but particular problems like anticipatory pleasure and giving trust in social situations. We could then build psychosocial treatments that work on those particular things.

BSJ: What are some treatments that you have developed for negative symptoms?

AK: One was a skills-based treatment – each week we would people teach a new skill. One skill was called “daily positives”. We asked people to pay attention to three positive things that happened in their day. It could be small things like getting a piece of chocolate when they weren't expecting it, or noticing a hummingbird while walking. The idea is to focus people's attention on positive things and link that up with their feelings. Another skill was called “reappraisal,” which is learning to think about a situation differently to change the way you feel about it. The goals in teaching these skills were to help with anticipatory pleasure as well as some social aspects of schizophrenia. An important lesson that we learned from this treatment was that over time, people forgot to use the skills. We can't go in and do an intervention and think we've solved the problem. For people with schizophrenia, it's going to require longer than six weeks. So we're working now to develop a treatment that lasts 15 or 16 weeks. After the treatment is over, we're going to do a booster session to remind people of the skills. More importantly, we're building mobile apps so we can incorporate the treatment into the course of daily life. One of my former students developed a mobile app to help people with motivation. It's a

"We're trying to harness the power of 21st-century technology to help people in their daily lives."

social media platform for people with schizophrenia to post their goals and accomplishments. Other people can respond with balloons, a thumbs-up, or other kinds of rewards. We're trying to harness the power of 21st-century technology to help people in their daily lives. In-person treatments are expensive, in terms of not just finances, but also time. If we could build treatments that we can start here and continue delivering them once the patient leaves, that will help with the effectiveness of the treatments.

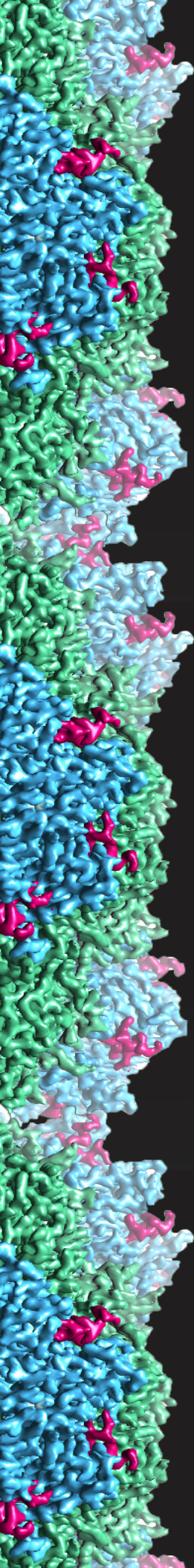
BSJ: What is some advice you would give to students who are interested in getting involved in research?

AK: I would say do it, because we need more science. For some reason, we're living in a society where people are starting to be skeptical of science, which I think is dangerous. I would say if somebody wants to get into the sciences, they absolutely should. It's a fascinating career, no matter what your field is. Whether you're in nanotechnology or psychology or anthropology or engineering, it's fascinating. If you like mysteries or puzzles, this is the career for you. You go in, and you've got an interesting hunch. So you turn a question into a hypothesis, and then you test it. It can be super rewarding. You can get to the bottom of the puzzle, and it's also fun because getting the answer helps you think of more questions. You have this never-ending curiosity because science raises as many questions as it answers.

BSJ: Thank you very much for your time!

REFERENCES:

1. Painter, J. M., & Kring, A. M. (2016). Toward an understanding of anticipatory pleasure deficits in schizophrenia: Memory, prospection, and emotion experience. *Journal of Abnormal Psychology*, 125(3), 442-452. doi:10.1037/abn0000151
2. Campellone, T. R., Fischer, A. J., & Kring, A. M. (2016). Using social outcomes to inform decision-making in schizophrenia: Relationships with symptoms and functioning. *Journal of Abnormal Psychology*, 125(2), 310-321. doi:10.1037/abn0000139



CRYO-ELECTRON MICROSCOPY: A CLOSER LOOK AT CYTOSKELETAL INTERACTIONS

Interview with Professor Eva Nogales

BY SHRUTHI CHOCKKALINGAM, ROSA LEE, WHITNEY LI, MELANIE RUSSO,
ELENA SLOBODYANYUK, SONA TRIKA

Dr. Eva Nogales is a Howard Hughes Medical Institute Investigator and Professor of Biochemistry, Biophysics, and Structural Biology at the University of California, Berkeley. Professor Nogales's research centers on using cryo-electron microscopy (cryo-EM) to investigate the mechanisms behind gene expression regulation and cytoskeletal dynamics in cell division. In this interview, we discuss the insights gleaned through the application of cryo-EM to study drug-stabilized microtubules.



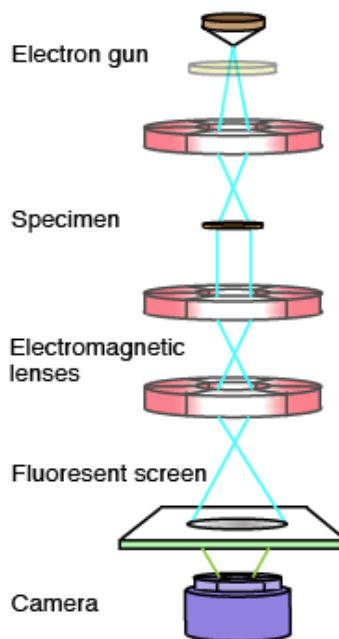
Professor Eva Nogales

BSJ: You originally studied solid-state physics. How did you get involved in the field of structural biology and cryo-EM?

EN: It was purely by chance. While I was an undergraduate in solid-state physics in Spain, I considered using synchrotron radiation techniques in the context of surface science. I met with the director of the British Synchrotron Radiation Source—he was a physicist and biologist, and a very charismatic individual, so I decided to join his research. The switch was just like that. It was very serendipitous, because the 21st century is the century of molecular biology, and I'm so glad to be in it. Once I made it to the synchrotron, I was using a technique called small-angle X-ray scattering to study tubulin self-assembly.

The technique was not that informative, so I started using cryo-EM to complement my studies with X-rays. It became obvious to me that cryo-EM is a very powerful technique. I came to Berkeley looking for postdoc positions and was referred to Kenneth Downing at Lawrence Berkeley National Laboratory. It was perfect—he was also a physicist, and because I had previous expertise with tubulin and he was very good at electron microscopy, we teamed up. But at the very beginning, it was purely accidental—moving from physics into biology and then getting exposed to cryo-EM at a time when the technique was just emerging.

Figure 1: Cryo-electron microscopy (cryo-EM) is a form of transmission electron microscopy (TEM) that allows the detailed observation of biological specimens at cryogenic temperatures, ranging from -180° to -269° C. In TEM, a thin layer of sample is hit with a high-energy beam of electrons. The electrons scatter onto a series of electromagnetic lenses, which produce a magnified 2D image of the specimen. The short wavelength of the electrons allows for sub-nanometer resolution in these images. [<http://cryoem.berkeley.edu/cryoem>]



BSJ: What are some advantages of cryo-EM over other conventional imaging technologies such as X-ray crystallography and nuclear magnetic resonance (NMR) spectroscopy?

EN: In cryo-EM, you don't need to crystallize your sample and you don't need a lot of it. You can also cope with conformational flexibility—in other words, you have a series of proteins that interact with one another in equilibrium, where they come together and go apart. This makes cryo-EM very generally applicable. That's the big benefit. In crystallography you need to be able to crystallize your sample, and in NMR you have to treat it with a special isotope. For determining protein structure *ab initio*, where you don't know what you're starting with, you're very limited in sample size for both techniques. Even for the average protein, the sample may be too small, while with cryo-EM you can study not just one protein, but an entire protein complex. Its advantage is its applicability, because it doesn't have all the requirements that other techniques have.

BSJ: A major factor in achieving near-atomic resolution was the advent of direct electron detectors in cryo-EM imaging. What is the advantage of direct electron detectors over charge-coupled detectors (CCDs)?

"With cryo-EM you can study not just one protein, but an entire protein complex."

EN: First, let me state something very important. Biological materials are radiation-sensitive—we cannot pass a lot of electrons through them because the sample will get damaged. But when we take images with very few electrons, it's like taking a photograph with very little light. As a result, the image looks blurry. In a CCD, the electrons enter a scintillator, where they are converted to light, which is then converted back to an electron signal to produce a digital image. In the process of an electron scattering in the scintillator, it goes from what used to be a dot to a cloud that hits the detector. As a result, the images are very noisy. Some engineering was necessary to remove the scintillator and make these detectors resistant to high-energy electron damage. Once this solution was obtained, the detector basically became noise-free. You end up with an image that has much better contrast and more signal per image than anything we were getting before. A second gain is that the read-out of these detectors is very fast. Instead of producing a single exposure, we are able to obtain a little movie in the same amount of time. We then add all the frames into a single image. With this, we are able to correct for beam-induced motion, which is caused by the water in the sample buckling upon being hit by electrons. We can do this trick because the images have more contrast and are collected faster. We went from having images where maybe five percent had high resolution and those images still had poor contrast, to where 100% of images have good contrast and high resolution. It's completely like night and day.

BSJ: Image analysis methodologies also play a key role in improving the resolution of cryo-EM images. Can you explain the difference between geometry-based approaches, cross-correlation, and maximum-likelihood (ML) methods?

EN: When you take an image in a transmission electron microscope, your image is a projection—the addition of densities along the direction of the electron beam. Image reconstruction involves using the projection to get a full 3D view of the object. When you take an image of a molecule, it is randomly oriented in solution. If you take those projections and identify their relative angles, you can combine them and achieve a 3D image of the object. Geometric principles are the simplest way to do this. You put your sample in the electron microscope, take an image, and then rotate it. You know exactly how the two images are related, because you did the rotation. Now imagine that you've used geometric principles to achieve the structure of a molecule, and you want to study that molecule bound to a drug. You can use your starting molecule as a reference, and computationally generate all possible projections from all angles. Then you compare your experimental images to the computer-generated ones with a mathematical procedure called cross-correlation. At the end of the day, you get the relative orientations of all your images with respect to a common reference. Why doesn't this work very well? Sometimes it's very hard to tell which image corresponds to which view. So here come ML methods. In ML, you don't make any strong deterministic assumption to start. Instead of saying "this image corresponds to that view," you just say, "there is a 50% chance that it corresponds to this view, but there is a 25% percent chance that it corresponds to that view, a 10% chance that it corresponds to that view, and one percent chance to each of these 15 other views." You calculate your reconstruction using

"Maximum likelihood methods not only allow you to identify the different orientations, but also the different conformations of a molecule."

weights, so that you don't commit the image to one view or the other. It's a soft assumption that plays with probabilities. It takes more computer time, but in the end, it converges better to the right solution. Additionally, ML methods not only allow you to identify the different orientations, but also the different conformations of a molecule. This is incredibly powerful, because we cannot assume that a protein is in just one state. Many protein complexes actually work by having parts that move. ML methods are much better at identifying the presence of these states in a sample and describing them in parallel to each other.

BSJ: What are some challenges you continue to face with cryo-EM?

EN: The biggest challenge is sample preparation. We have a protein, an organic molecule that is made of carbon, nitrogen and oxygen. It is surrounded by water, which also has oxygen. The contrast in the image is given by different scattering of electrons, and if the background is as dense as your molecule, you don't see anything. You can minimize the amount of water in the sample, but you have an air-water interface, which is a large hydrophobic surface. If a protein has a very hydrophobic section, it is perfectly happy to unfold and interact with the air-water interface. Right now, the process of thinning the sample and keeping it happy is hit-or-miss. We use tricks, cross-linkers that keep the protein from unraveling. But sample preparation is our bottleneck. That, along with access to electron microscopes. I wish we had an electron microscope that anybody could use at any time. But we have to share it, for many projects and many labs.

BSJ: We read about some of your work imaging microtubules.¹ What is the seam and what role does it play in the structure of the microtubule?

EN: Tubulin has this amazing property of self-association, meaning that it forms a polymer by itself. Tubulin dimers interact head-to-tail to form linear protofilaments, which then associate laterally to close into a tube. Most of the contacts are beta-to-beta and alpha-to-alpha, except at one point where beta interacts with alpha and alpha interacts with beta. This is the seam. What function does it have? We actually don't know, but we think it's involved in the process of microtubule closure, a zipping-up process in which many contacts are made simultaneously. It could also be a weak point where contacts are not very optimal, and that's what allows the microtubule to break down in dynamic instability.

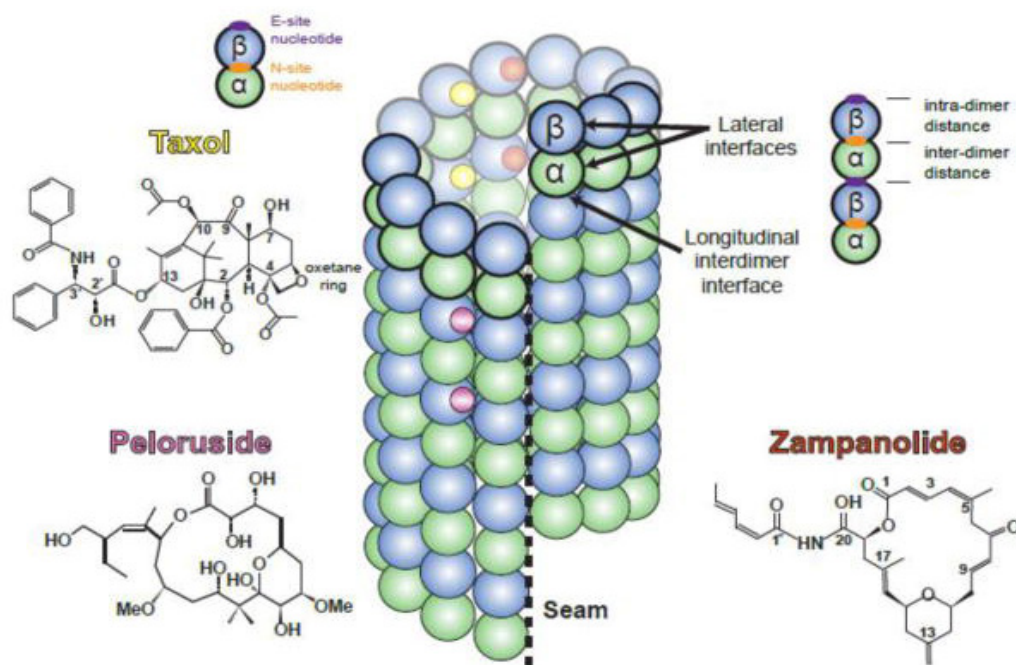


Figure 2: Schematic of a microtubule featuring the seam and the three binding sites for antimitotic drugs Taxol, peloruside, and zampanolide.²

BSJ: What is the property of dynamic instability in microtubules? What are microtubule-stabilizing agents and how do they relate to this phenomenon?

EN: Microtubules are not rigid. If you fluorescently label a microtubule and see how it behaves in real time, you see that it grows for a while and then disassembles—this phenomenon is called dynamic instability. It is very weird behavior for a polymer. In the process of cell division, microtubules grab onto chromosomes and pull and push on them until they are aligned. Then the two chromatids split apart and the microtubules pull them to each side of the cell. In order for microtubules to go through all of this growing and shrinking, they need to be dynamic. If that is stopped—if you give the cell a drug that makes the microtubule very stable, for example—it gets stuck in cell division and commits apoptosis, or cell death. That's how these antimitotic drugs work in cancer cells.

BSJ: You have investigated a variety of antimitotic cancer drugs, including Taxol, peloruside, and zampanolide.² What motivated your selection of these drugs to investigate?

EN: Taxol is the most famous and more broadly-used of the antimitotic drugs. In fact, it's used in the treatment of almost every single solid tumor. Therapeutically, it is the most important antimitotic

drug. Zampanolide is interesting because it binds to the same pocket as Taxol, but it binds covalently. We wanted to see whether these two drugs, which bind to the same place, have the same effect on the microtubule. Peloruside binds in a completely different place, but it also stabilizes the microtubule. The three drugs were chosen because we wanted to know whether they stabilize the microtubule in a similar or different way.

BSJ: How do Taxol-site binders such as Taxol and zampanolide affect microtubule structure? How does this compare with the effect of non-Taxol-site binders, such as peloruside?

EN: What we see with Taxol and zampanolide is that they cause the microtubule lattice to be flexible, so that there is variability in the lateral contacts between protofilaments. The microtubule is wiggling. It's exactly the opposite for peloruside. Peloruside binds, bridging two protofilaments, and it makes the microtubule structure very ordered, so the opposite of wiggling—very well-defined lateral contacts between all the protofilaments.

BSJ: What is a doubly bound microtubule? What did you find in your investigation of the effect of a peloruside-Taxol doubly bound structure on microtubule structure?

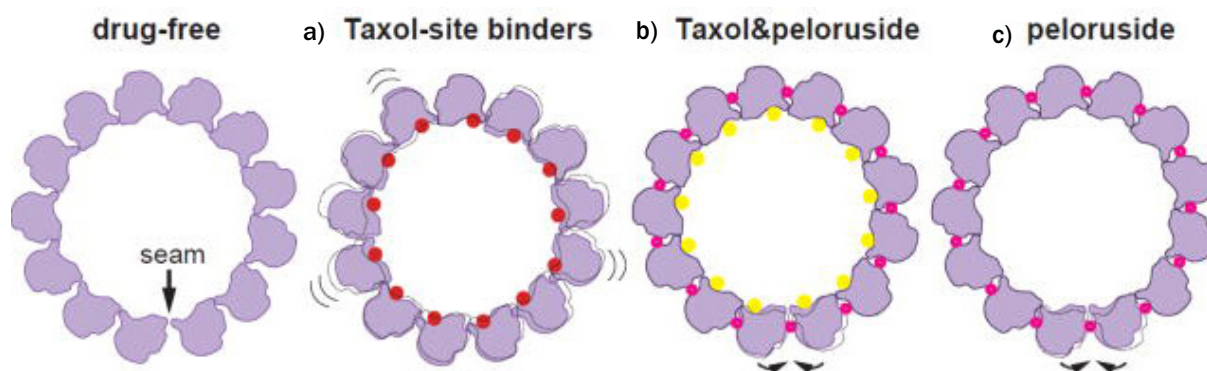


Figure 3. (a) Taxol-site binders Taxol and zampanolide causes the microtubule lattice to be flexible. (b) In a doubly-bound microtubule, peloruside overrides the effects of Taxol, making the microtubule lattice more ordered. (c) Peloruside promotes order in the microtubule lattice.²

EN: Since peloruside binds on one site, and Taxol binds on the other, we can add both agents to the microtubule because they don't compete with each other. This is interesting because you can conceive of the possibility of treating cancer by using a combination of both drugs. Because they have such different effects on the microtubule lattice, we decided to bind both of them and see their effects on microtubule structure. What we find is when both peloruside and Taxol are bound, the lattice becomes very well-ordered, so the effect of peloruside on lattice contacts wins over that of Taxol. We think that Taxol has more of an effect on longitudinal interactions (interactions between subunits along the length of the protofilament), while peloruside favors contacts between protofilaments. Both stabilization effects are additive, and, in terms of lattice order, peloruside makes even Taxol-bound microtubules well-ordered.

BSJ: Thank you very much for your time!

REFERENCES

1. Nogales, E., & Kellogg E. H. (2017). Challenges and opportunities in the high-resolution cryo-EM visualization of microtubules and their binding partners. *Current Opinion in Structural Biology*, 46, 65-70. doi:10.1016/j.sbi.2017.06.003
2. Kellogg, E. H., Hejab, N. M. A., Howes, S., Northcote, P., Miller, J. H., Diaz, J. F., Downing, K. H., Nogales, E. (2017). Insights into the distinct mechanisms of action of taxane and non-taxane microtubule stabilizers from cryo-EM structures. *Journal of Molecular Biology*, 429(5), 633-646. doi:10.1016/j.jmb.2017.01.001

BSJ: What are some future directions of your research?

EN: Concerning microtubules, we are very interested in studying how different cellular factors affect microtubule behavior, especially dynamic instability. Many different factors can interact with microtubules at different states during the cell cycle or in different tissues. For example, we are now studying one microtubule-associated protein called tau that binds to axons. We are interested in investigating how tau stabilizes those microtubules and favors axonal regular function. We are also very interested in how microtubules interact with chromosomes via kinetochore complexes that can link together the microtubule and special nucleosomes in chromatin. These are both areas of research that are of great interest to us—cellular control of microtubule dynamics and the engagement of microtubules with chromosomes for the segregation of genetic material during cell division.